The Astronomical Journal, 152:111 (24pp), 2016 November doi:10.3847/0004-6256/152/5/111
© 2016. The American Astronomical Society. All rights reserved.

A CAREFULLY CHARACTERIZED AND TRACKED TRANS-NEPTUNIAN SURVEY: THE SIZE DISTRIBUTION OF THE PLUTINOS AND THE NUMBER OF NEPTUNIAN TROJANS

Mike Alexandersen[1,2], Brett Gladman[1], J. J. Kavelaars[3], Jean-Marc Petit[4], Stephen D. J. Gwyn[3], Cory J. Shankman[5], and Rosemary E. Pike[2,5]
[1] Department of Physics and Astronomy, 6224 Agricultural Road, University of British Columbia, Vancouver, BC V6T 1Z1, Canada; mike.alexandersen@alumni.ubc.ca
[2] Institute of Astronomy and Astrophysics, Academia Sinica, Taipei, Taiwan, R.O.C.
[3] National Research Council of Canada, Victoria, BC V9E 2E7, Canada
[4] Institut UTINAM, CNRS-UMR 6213, Observatoire de Besançon, BP 1615, F-25010 Besançon Cedex, France
[5] Department of Physics and Astronomy, University of Victoria, Victoria, BC, Canada
Received 2014 November 27; revised 2016 May 19; accepted 2016 June 7; published 2016 October 7

ABSTRACT

The trans-Neptunian objects (TNOs) preserve evidence of planet building processes in their orbital and size distributions. While all populations show steep size distributions for large objects, a relative deficit of Neptunian trojans and scattering objects with diameters of $D < 100$ km has been detected. We investigated this deficit with a 32 square degree survey, in which we detected 77 TNOs that are brighter than a limiting r-band magnitude of 24.6. Our plutino sample (18 objects in 3:2 mean-motion resonance with Neptune) shows a deficit of $D < 100$ km objects, rejecting a single power-law size distribution at >99% confidence. Combining our survey with the Canada–France Ecliptic Plane Survey, we perform a detailed analysis of the allowable parameters for the plutino size distribution, including knees and divots. We surmise the existence of 9000 ± 3000 plutinos with an absolute magnitude of $H_r \leqslant 8.66$ and 37000^{+12000}_{-10000} with $H_r \leqslant 10.0$ (95% confidence). Our survey also discovered one temporary Uranian trojan, one temporary Neptunian trojan, and one stable Neptunian trojan, for which we estimate populations of 110^{+500}_{-100}, 210^{+900}_{-200}, and 150^{+600}_{-140} with $H_r \leqslant 10.0$, respectively. All three populations are thus less numerous than the main belt asteroids (592 asteroids with $H_r \leqslant 10.0$). With such population sizes, the temporary Neptunian trojans cannot be previously stable trojans diffusing out of the resonance now; they must be recently captured Centaurs or scattering objects. As the bias against the detection of objects grows with larger semimajor axes, our discovery of three 3:1 resonators and one 4:1 resonator adds to the growing evidence that the high-order resonances are far more populated than is typically predicted.

Key words: astrometry – Kuiper belt: general – methods: data analysis – methods: observational – planets and satellites: detection – surveys

Supporting material: machine-readable table, tar.gz files

1. INTRODUCTION

The outer Solar System (beyond ~5 au) contains four giant planets, thousands of Jovian trojans, at least four dwarf planets with diameters $D > 1000$ km, and thousands of smaller trans-Neptunian objects (TNOs) whose numbers generally rise dramatically with decreasing size. Because the sizes of TNOs are not directly measurable, the best proxy for studying the size distribution is the absolute-magnitude distribution. These are equivalent if one assumes all TNOs have the same albedo (reflectivity). Minor body populations can typically be well represented over some range of absolute magnitudes by an exponential absolute-magnitude distribution. The distribution has the differential form $\frac{dN}{dH} \propto 10^{\alpha H}$, where dN/dH is the number of objects per H-mag, and the exponent α (and the constant of proportionality) is specific to the dynamical population of interest. Assuming a single albedo value for all objects and converting H-magnitude to diameter, the exponential magnitude distribution corresponds to a power-law diameter distribution $dN/dD \propto D^{-\mathfrak{q}}$, where $\mathfrak{q} = 5\alpha + 1$ is the differential size index (not to be confused with q, the distance of pericenter).

The TNO populations have been shown to feature steep ($\mathfrak{q} \gtrsim 4$, $\alpha \gtrsim 0.6$) distributions for $D > 100$ km (Jewitt et al. 1998; Gladman et al. 2001; Bernstein et al. 2004; Elliot et al. 2005; Petit et al. 2006; Fraser & Kavelaars 2008; Fuentes & Holman 2008). These steep distributions are thought to be imprints of planet accretion, as $D < 10$ km objects accumulate into larger objects through collisions, which is known as "bottom-up formation" (Kenyon & Luu 1998; Weidenschilling et al. 1997). Kenyon & Luu (1999) numerically investigate many different scenarios in which they simulate collisions of a primordial disk of sub-km objects. Despite using various different parameters for the typical impact velocity and likelihood of merger (versus catastrophic destruction), they find that most of their simulations produce large planetesimals with a power-law size distribution with $\mathfrak{q} \approx 4$ ($\alpha \approx 0.6$) in the 1–500 km range. Similar numerical accretion calculations by others often produce $\mathfrak{q} \approx 4$ (Kenyon & Luu 1998; Schlichting et al. 2013). Analytic work, such as that by Dohnanyi (1969) and O'Brien & Greenberg (2003), conclude the equilibrium index to be $\mathfrak{q} = 3.5$ and $3.04 \leqslant \mathfrak{q} \leqslant 3.66$ ($\alpha = 0.5$ and $0.408 \leqslant \alpha \leqslant 0.532$), respectively; however, both of these analytic works make simplifying assumptions about the dependence of object strength with size (Dohnanyi 1969 assumes no dependence). Observationally, even steeper distributions ($\mathfrak{q} = 5$–7, $\alpha = 0.8$–1.2) are measured for $D > 100$ km TNOs (Gladman et al. 2001; Bernstein et al. 2004; Elliot et al. 2005; Petit et al. 2011; Adams et al. 2014). Other Solar System populations, such as the asteroid belt and the Jovian trojans, also feature steep

distributions with $\mathfrak{q} = 4.5$ ($\alpha = 0.7$) and $\mathfrak{q} = 5.5 \pm 0.9$ ($\alpha = 0.9 \pm 0.2$) for $D \gtrsim 100$ km, respectively (Jewitt et al. 2000; Bottke et al. 2005), while $D < 100$ km main belt asteroids show $2.2 \leqslant \mathfrak{q} \leqslant 4.1$ ($0.24 \leqslant \alpha \leqslant 0.62$; Jedicke et al. 2002, p. 71; Gladman et al. 2009b). A likely explanation for how these populations (other than the $D < 100$ km main belt asteroids) can have size distributions that disagree with the predicted equilibrium state is that the populations are not, and never have been, in accretional/collisional equilibrium. The populations may have formed quickly with a steep size distribution, after which planetary migration scattered the planetesimals, lowering the spatial density such that the populations are too collisionally inactive to have reached collisional equilibrium.

While the diameter distribution is the real physical property, the albedo is often unknown and thus the sizes of observed objects are unknown; only their absolute magnitude is known. We therefore exclusively use the absolute-magnitude distribution for analysis in this work. Assuming a TNO r-band albedo of 5% (as used by Sheppard & Trujillo 2010b; Petit et al. 2011; Gladman et al. 2012, although this is in the low end of recently measured TNO albedos from Mommert et al. 2012; Lacerda et al. 2014), $D \simeq 100$ km is equivalent to an absolute r-band magnitude of $H_r = 8.9$. This was calculated using a conversion similar to that used by Ivezić et al. (2001):

$$H_r = m_{\mathrm{Sun},r} - 2.512 \log\left(\frac{p_r \times D^2}{4 \times (1\ \mathrm{au})^2}\right) \quad (1)$$

where p_r is the r-band albedo, D is the diameter, and $m_{\mathrm{Sun},r} = -26.91 \pm 0.02$ is the apparent magnitude of the Sun, which was calculated using

$$r = V - 0.46 * (B - V) + 0.11 \quad (2)$$

from Jester et al. (2005), with $V_{\mathrm{Sun}} = -26.75$ (Allen 1973) and $(B - V)_{\mathrm{Sun}} = 0.642 \pm 0.016$ (Holmberg et al. 2006).

The size distribution of observationally accessible 100 km $< D <$ 1000 km TNOs has been the subject of intense study (e.g., Fuentes & Holman 2008; Petit et al. 2008, p. 71), while $D < 100$ km has been studied less due to the need to probe moving targets with apparent magnitudes $m_r \gtrsim 24$. Interpreting these results is complicated by the realization that multiple subpopulations are present with different magnitude and orbital distributions (Bernstein et al. 2004; Elliot et al. 2005; Fraser et al. 2010; Petit et al. 2011). Fuentes & Holman (2008) investigated the luminosity function of TNOs as a whole, and found that a single exponential cumulative magnitude distribution could not continue past $m_r \sim 25$, determining a bright object slope of $\alpha_b = 0.75^{+0.12}_{-0.08}$, with a transition at $m_r = 24.9^{+0.5}_{-0.9}$ to a faint slope of $\alpha_f = 0.23^{+0.07}_{-0.14}$. However, they point out that the trans-Neptunian region is made up of multiple populations. Splitting their sample into a "classical" and "excited" subsample, they find that while the two populations independently reject a single exponential, they favor significantly different slopes and transition parameters. Of course, investigating the luminosity function rather than the absolute-magnitude distribution means that any features of the distribution will be smeared out because the objects are at a range of distances.

Kenyon et al. (2008, p. 293) perform accretion models while simultaneously trying to account for how dynamically excited the trans-Neptunian region is, compared to flat, low-eccentricity disks that are usually used in accretion simulations. They find that once 10–100 objects with $D \sim 600$–2000 km are produced, accounting for ~10% of the disk mass, these bodies will excite the remaining disk (raising the average collision speed) sufficiently to halt accretion. This causes the small objects to reach collisional disruption equilibrium with $\mathfrak{q} = 3.5$ ($\alpha = 0.5$), while large objects retain $\mathfrak{q} = 2.7$–4 ($\alpha = 0.34$–0.6). While the transition size is model-parameter dependent, they find the transition between the accretion and disruption slopes to be $D \sim 40$–80 km. A transition in the size distribution could thus be produced by processes as described in Kenyon et al. (2008, p. 293).

Recent work with more precise distance measurements has found a dramatic change in the magnitude distribution of Neptunian trojans (Sheppard & Trujillo 2010b) and scattering objects (Shankman et al. 2013), with a significant lack of $D < 100$ km objects (relative to an extrapolation of the exponential distribution of larger objects). This might indicate that planetesimals were "born big" (Morbidelli et al. 2009), skipping the intermediate sizes, which is in sharp contrast with bottom-up formation; this may explain how $\alpha > 0.6$ for $D > 100$ km. In the born-big scenario, any intermediate-sized objects seen today are collisional fragments. Objects with $D < 10$ km accrete to $D > 100$ km objects rapidly, avoiding the creation of $D < 100$ km objects, through unknown processes. The $D > 100$ km objects then collisionally evolve to produce $D < 100$ km objects, forming a divot in the size distribution; if the population is collisionally active for sufficiently long, the $D < 100$ km dearth can be completely erased (Campo Bagatin & Benavidez 2012). An alternative explanation for the apparent lack of $D < 100$ km objects could be that a sudden change in the typical albedo of TNOs occurs at this size. However, the sudden lack of Neptunian trojans seen by Sheppard & Trujillo (2010b) would require a sudden decrease in the average albedo by a factor of 6. Observational measurements of TNO albedos (by occultations and observations of binary TNOs) have revealed that the TNOs have a wide range of albedos, and the average albedo does not appear to be size-dependent (Grundy et al. 2005). So while a size-dependent albedo distribution with a very sudden transition could explain the observed lack of objects, it appears less plausible than a transition in the size distribution.

Around 1030 TNOs are known (with $a > a_{\mathrm{Neptune}}$ and observations in multiple years).[6] However most of them were discovered in surveys that have not been "well-characterized" (as defined in Jones et al. 2006) or have not had the characterization published, so the biases involved in the discoveries are not well-known. Survey characterizations are important to perform accurate modeling of the observations. The Canada–France Ecliptic Plane Survey (CFEPS) is one of the largest TNO surveys, and has published its characterization (Petit et al. 2011). However, CFEPS had very little sensitivity to objects with $D < 100$ km. To improve upon the conclusions of CFEPS and to investigate the lack of $D < 100$ km objects described above, we performed a deeper survey designed to probe the faint objects beyond the sensitivity limit of CFEPS.

Several resonant populations may reach pericenter inside Neptune's orbit while still being long-term stable. It is thus possible to detect smaller resonant TNOs than classical TNOs.

[6] List of distant Solar System objects in the Minor Planet Center (MPC) database as of 2015 March 24: http://web.archive.org/web/20150324222059/http://www.minorplanetcenter.net/iau/MPCORB/Distant.txt.

Our survey was specifically designed to probe the plutinos and the Neptunian trojans, because the plutinos are very numerous and come to pericenters near the trojan clouds. We were able to probe to $D < 100$ km within these populations. Our goal was to confirm or refute the drastic drop in number density seen by Sheppard & Trujillo (2010b) for $D < 100$ km Neptunian trojans. We also aimed to investigate whether such a drop in object density could be confirmed or rejected for the plutinos. The plutinos and Neptunian trojans may have had a similar origin (Levison et al. 2008; Lykawka et al. 2011), so the plutinos, being vastly more populous, could provide even stronger evidence for or against such a paucity.

There are several suggested absolute-magnitude distribution shapes in the literature aside from the single parameter exponential. Fuentes & Holman (2008) and Fraser et al. (2014) suggest a three parameter broken exponential (a "knee") for hot populations. For the scattering objects, Shankman et al. (2013) proposes a four parameter "divot" (two exponentials with a drop at the transition); a sufficiently deep divot may also explain the lack of $D < 100$ km Neptunian trojans seen by Sheppard & Trujillo (2010b). We thus investigated the single exponential, knee, and divot scenarios, and explored which model parameters are compatible with our data.

Sections 2 and 3 describe the technical details of the survey, including the survey design and methodology, concluding with a summary of our detections. Detailed modeling of the plutinos is presented in Section 4; this section shows that the plutinos from this survey reject a single exponential size distribution, and also combines our survey with CFEPS for a detailed analysis of allowable parameters. We also use the combined surveys to calculate population estimates for Neptunian and Uranian co-orbitals (Section 5) and for the 3:1 and 4:1 mean-motion resonances with Neptune (Section 6).

2. OBSERVING STRATEGY

Our survey was designed to obtain 40 sq.deg. of high-cadence sky coverage near the right ascension (R.A.) of 2 hr. Despite the severe weather and technical problems in 2011 and 2012, we succeeded in obtaining high-quality observations for 32 sq.deg., thanks to a robust survey design. The survey was conducted using MegaCam on the Canada–France–Hawaii Telescope (CFHT) on Maunakea, Hawaii, which was chosen for its combination of a large field of view (almost $1^\circ \times 1^\circ$) and a 3.58 m diameter aperture, providing the best combination of depth and coverage. Additionally, the queue-operation of CFHT was crucial for the success of this survey, which required many observations throughout two semesters, but often only a couple of hours of observations in a given night.

The on-sky pointings of the survey were chosen to be near Neptune's L4 Lagrange point, where many Neptunian trojans reside and many n:2, n:3, and n:4 resonances are near pericenter (Gladman et al. 2012). The survey obtained one "low-lat" block of 20 sq.deg. ($5^\circ \times 4^\circ$) centered near the ecliptic plane and another "high-lat" block of 12 sq.deg. ($4^\circ \times 3^\circ$) located $\sim 15^\circ$ north (see Figure 1 for block geometry and Table 1 for exact coordinates). We divided our survey this way to better constrain the inclination distribution of the populations. The plutinos and Neptunian trojans are dynamically hot populations, so their inclination distributions are wide, with characteristic widths around $\sim 15^\circ$ (Gladman et al. 2012; Parker 2015). Ecliptic surveys are strongly biased against detecting high-inclination objects, because these objects spend more time well away from the ecliptic. Even with the decrease in the total number of detections, a survey at higher latitudes should detect more high-inclination objects than an equivalent ecliptic survey (Lawler & Gladman 2013), providing a better lever arm for statistical tests of the extent of the inclination distribution of the populations.

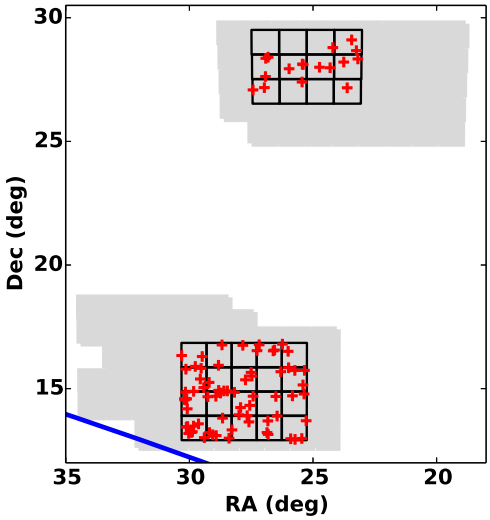


Figure 1. Sky coverage of our survey. Black boxes are the 1 sq.deg. field of view of CFHT's MegaCam, showing our discovery fields; crosses are the location of our detected objects at the time of discovery; the inclined blue line in the lower left is the ecliptic. The shaded gray areas are the "wallpaper" coverage described in Section 2.1.

Our survey needed to reach $m_r \approx 24.5$ to detect objects with $H_r \approx 9.7$ at 30 au. This depth is more than a magnitude past the reported transition (Sheppard & Trujillo 2010b; Shankman et al. 2013) near $H_r \approx 8.5 \pm 0.2$. Our survey would detect 3–5 Neptunian trojans if the sky density reported in Sheppard & Trujillo (2010b) was representative, or 20–30 if the lack of small objects reported in that work was a statistical fluke and the steep ($\alpha \approx 0.8$) distribution continues.

The discovery observations for each field were a set of three 320 s exposures (plus 40 s overhead) separated by $\sim$1 hr ("triplets"); 10 fields could thus obtain discovery images in a three-hour block. The number of fields observed at once (and thus the time between images in a triplet) had to be large enough that the objects would move appreciably from image to image, but we were also limited by the fact that we required good steady image quality and photometric conditions for the duration of the triplets. With typical image quality of $\sim 0\farcs5$–$0\farcs7$, the lower limit of detectable motion is $\sim 0\farcs3$/hour, which is equivalent to an object at $\sim$500 au. However, such an object would need to be significantly larger than Pluto or Sedna to be detected in our survey.

Our fields obtained discovery observations in 2011 and 2012 October. Our "low-lat" block was divided horizontally into two 10 sq.deg. sub-blocks obtaining triplets, as described above. The southern half obtained discovery observations on 2011 October 24 and the northern half on 2011 October 26, with image qualities mostly in the range $0\farcs5$–$0\farcs6$. The high-lat block obtained discovery triplets in 2012 October, with image qualities ranging from $0\farcs5$ to $0\farcs8$. The 2012 October high-lat discovery block was divided vertically into two 6 sq.deg. sub-blocks. The discovery triplets consisted of the six fields of the sub-block, interspersed with single images ("nailing" images) of fields from the other sub-block to extend the interval

Table 1
Field Coverage of Our Survey

Width (°)	Height (°)	R.A.	Decl.	JD	Fill Factor (%)
			Low-latitude block		
4.930	0.995	01:51:08.00	+13:24:48.0	2455858.963195	90.9
4.914	0.995	01:51:08.00	+14:23:36.0	2455858.963195	91.2
4.896	0.995	01:51:08.00	+15:22:24.0	2455860.843750	91.5
4.878	0.995	01:51:08.00	+16:21:12.0	2455860.843750	91.6
			High-latitude block		
1.993	2.984	01:41:01.70	+28:10:00.0	2456220.904167	90.7
1.993	2.984	01:49:58.30	+28:10:00.0	2456221.947917	90.7

Note. The width and height describe a rectangle centered on the given R.A. and decl. JD is the Julian date of the discovery observations. The fill factor is the fraction of the area inside the rectangle that was covered (less than 1 due to chip gaps and separation of fields).

between subsequent images of the triplets up to near an hour. The western half obtained discovery observations on 2012 October 20; the eastern half was observed on 2012 October 21.

Ideally, both blocks would have obtained discovery images at opposition in the October dark run of 2011, because the rate of motion is highest at opposition and the objects are brightest. Poor weather and a short October dark run meant that our "high-lat" block did not obtain any imaging in 2011 October. Our attempts to obtain (less ideal) discovery triplets in August and December were also thwarted by poor weather and technical difficulties. As data delivery, data reduction, characterization of the observations, and identification of objects takes several months, we did not plan to perform targeted recoveries within the discovery year. Our blocks were large contiguous fields, and thus shifting them across the sky at the mean TNO rate during the semester allowed us to successfully recover most objects in several other dark runs, with very limited loss due to the Keplerian shear. Our high-lat observations from 2011 August, 2011 November, and 2012 January were thus used as precovery images for the 2012 discovery observations. By reducing the $5^\circ \times 4^\circ$ grid used in 2011 to a $4^\circ \times 3^\circ$ grid for the 2012 high-lat discoveries, we ensured that the preemptive follow-up block was large enough to accommodate the shearing of objects discovered in 2012. This approach allowed us to complete the project with a high tracking fraction within two years, although at a 20% reduction in total field coverage.

Objects were tracked for up to 28 months of total arc. Single "nailing" images were taken at various times to extend the arc of the astrometry. At least one nailing image is required in the discovery dark run, although two is preferable to prevent losing objects in chip gaps. In the discovery year (both years for the high-lat block), nailings for tracking the objects outside the discovery dark run were performed by simply shifting the entire contiguous block across the sky at the mean TNO rate (see Figure 1). After each dark run in 2012, newly obtained images were searched manually (by eye) for our real objects. This process was aided by a software package developed by J. J. Kavelaars, which significantly improved throughout our use on this project. That software calculates the predicted location for the times of the new observations, and displays any images that cover the predicted locations, with overlays showing the predicted on-sky position and uncertainty. Recovery was usually trivial in cases where multiple nailings of a field were obtained within a dark run.

Once an object was located in an image near the discovery night, the improved prediction (based on the supplemented astrometry) was used to locate the object on further images. Once the observed arc was a few days long, the on-sky uncertainty was usually sufficiently small for the object to be easily located on images from one month before or after discovery. This iterative process continued until the object was found on all available images, or the remaining images were deemed unusable. Without observations outside the discovery dark run, the objects would have been impossible to locate manually a year later, and recovery would have required searching new triplets.

By the end of the discovery year, most objects had several months of well-sampled arcs. The on-sky predicted locations for the following/previous years were therefore accurate to a few arcseconds. For the low-lat block, pointed recoveries in 2012B were trivial due to the small on-sky uncertainty of most objects, saving ≈30% of telescope time compared with a large contiguous block, while also having a higher recovery rate because some objects had sheared up to a couple of degrees from the average motion.

2.1. Accurate Astrometry

The uncertainty in the heliocentric orbits of a TNO depends on the observed arc, the number and temporal distribution of observations inside the arc, and the accuracy of the measured astrometry. Larger astrometric uncertainty allows a wide range of acceptable orbital fits, and astrometry affected by systematic errors pushes the determined orbit away from the true parameters. Regular recovery over long arcs can help counter these problems, but obtaining the best possible orbit determinations in a short amount of time requires the best possible astrometry. A particular concern with short arcs is systematic errors, which can introduce errors of order ∼0.2 arcseconds when the astrometric references have systematic offsets. To limit systematic errors in our astrometry as much as possible, two large plate solutions and stellar catalogs were made (one for each block), which served as block-wide astrometric reference catalogs.

To make the unified astrometric catalog, every image from 2011 to 2012 was stitched together. The slow drift of fields across the sky produced considerable overlap, allowing the fields to be stitched together and chip gaps to be filled. To ensure that all portions of the sky coverage were connected (including outlying fields from pointed recoveries in 2012), we also observed two grids of shallow, 20 s exposures, which were

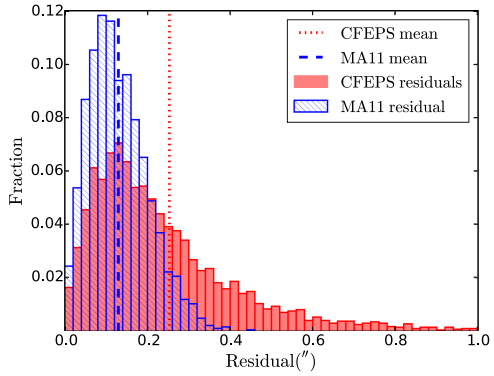


Figure 2. All residuals for orbit fits to astrometry from CFEPS (red) and this survey (MA11, blue), with the survey means shown with vertical lines. Residuals are the x and y residuals combined in quadrature.

used to "wallpaper" all our other pointings. The two grids were offset by half a chip in the vertical direction and 1.5 chips in the horizontal direction to ensure overlap between neighboring fields and to remove the distortion of the focal-plane. The coverage of the wallpaper, and thus the extent of our astrometric catalog, can be seen in Figure 1.

The astrometry of each MegaCam image, both the science image and the wallpaper images, was individually calibrated using the 2MASS catalog (Skrutskie et al. 2006) as a reference. Catalogs containing R.A., decl., and magnitude were created for each image. These catalogs were merged into a single catalog, and sources found in multiple images were identified and their positions were averaged together. This merged catalog was used to calibrate all the images. The resulting astrometric calibration has internal uncertainties on the order of 0.04 arcsec. Our final astrometric catalog is available in the supplementary files.

This survey-wide astrometric catalog allowed us to measure accurate astrometry over a two-year period, leading to unprecedented small residuals from our orbit fits, with mean residuals of $0\farcs13$ and a maximum residual of $0\farcs45$ (where x and y residuals were combined in quadrature). These residuals are dominated by measurement uncertainties in the positions of the objects themselves (due to a low signal-to-noise ratio), not by uncertainties in the astrometric catalog. This is a significant improvement over the $0\farcs25$ rms residuals of CFEPS (see Figure 2), providing significantly better orbit determinations with a short arc. This improvement allowed us to perform pointed recovery observations of almost all low-lat objects in 2012B, based on just six-month arcs in 2011B, and to securely classify most objects at the end of 2012B (see Section 3.4).

3. CHARACTERIZATION METHODS

The triplets from the discovery night were passed through the moving-object pipeline developed by the CFEPS team (Petit et al. 2004). This pipeline employs two different moving-object detection algorithms to search for linearly moving sources: one using Source Extractor and the other using a wavelet transform method for detecting point sources. Each algorithm produces a large number of false candidates, mostly caused by the pipeline detecting random noise, scattered light, cosmic rays, or faint background sources that only get detected in one or two images due to variable seeing. The different nature of source detection in the two algorithms means that the set of false candidates in each has only a small overlap. A large fraction of false candidates were removed by automatically rejecting any candidates that are were not identified by both algorithms. Only the remaining candidates were subsequently vetted by human inspection.

To determine the characteristics of our survey (detection efficiency and false-positive rate) we planted artificial objects using methods outlined in Petit et al. (2004) and Jones et al. (2006). Here we present a new improvement to this process. We did not plant the artificial objects into the "real" triplets that were searched for real objects. Instead we used a copy of the triplet images, which we first temporally scrambled; that is, we switched around the UT timestamps in the image headers so that images 1, 2, 3 were permuted to the sequence 2, 3, 1. This scrambling meant that any detections by the pipeline in these "fake" image triplets must either be artificially implanted objects or false positives; they cannot be real objects, because no real object could move linearly in these out-of-order images. That is, no real outer Solar System object can reverse its apparent direction of sky motion within two hours. To our knowledge, this approach, which allows us to measure the false-positive rate exactly, has never been used before. This type of science (with short-term monotonic time variability) is uniquely able to perform such false-positive measurements.

After passing the real and fake triplets through the automated pipeline, a human operator inspected every candidate from the real and fake sets simultaneously. The candidates were mixed together, such that the operator never knew whether a candidate from a real or fake triplet was being shown. The blind inspection ensured that the two sets were given the exact same treatment.

During the human inspection, detections that were believed to be invalid (mostly caused by noise and stellar diffraction spikes) were rejected by the operator. The operator also assigned flags to observations, flagging images where the astrometry or photometry might not be trustworthy. This included images with stellar interference in the object's point-spread function or bright sources that were significantly polluting the photometric sky annulus.

Once all candidates were either approved or rejected, the approved objects in the fake set of triplets were used to determine the efficiency function and the false-positive rate. The approved objects in the real set of triplets are real objects (or false positives) and were subsequently searched for in tracking observations.

3.1. Efficiency of Search

The detection efficiency was calculated as a function of magnitude, for a few rate of motion ranges, to investigate whether our efficiency varied significantly with rate. The efficiencies as functions of magnitude (for $m_r > 21$) were found to be well represented by a function described in Gladman et al. (2009b), of the form:

$$f(m_r) = \frac{f_{21} - k(m_r - 21)^2}{1 + \exp\left(\frac{m_r - m_e}{w}\right)}, \quad (3)$$

where f_{21} is the efficiency at $m_r \simeq 21$, k is a measure for the strength of a quadratic drop (caused by crowding; objects being obscured by background stars is more problematic for fainter objects), m_e is the magnitude at which the function transitions to being an exponential tail, and w is a measure for the width of that tail.

Table 2
Efficiency Function Parameters and Characterization Limits of Our Survey Blocks in the Searched Rate Ranges

Rate Range ("/hr)	f_{21}	k	m_e	w	Limit
Low-latitude block					
0.50–2.06	0.96	0.0148	24.62	0.125	24.62
2.06–3.72	0.91	0.0118	24.60	0.133	24.61
3.72–5.38	0.90	0.0106	24.55	0.142	24.57
5.38–10.00	0.88	0.0136	24.48	0.130	24.49
High-latitude block					
0.50–2.06	0.92	0.0194	24.64	0.118	24.60
2.06–6.00	0.90	0.0133	24.59	0.115	24.60
6.00–10.36	0.88	0.0139	24.49	0.124	24.49

Note. The efficiency function is parameterized in Equation (3). The limit is the magnitude at which the efficiency hits 40% of the maximum value. The functions are plotted in Figure 3.

The limiting magnitude of the survey for each rate range was chosen to be the magnitude at which the efficiency was 40% of the maximum efficiency, rounded up to nearest 0.01 mag. Table 2 gives the limiting magnitudes for the blocks and the parameters used to model the efficiency functions, which are also shown graphically in Figure 3.

3.2. False-positive Rate

The fake triplets had 22,910 candidate detections, of which 1052 were rejected and 21,858 were accepted; of the accepted objects, 21,855 were planted, leaving only three false positives (i.e., the human operator was willing to accept these three detections as real candidates near the noise limit, although they were actually coincidental noise). Having only three false positives despite the sheer number of candidates shows the ability of humans to successfully filter noise from valid low signal-to-noise sources. Moreover, these three false positives were all beyond our survey's characterization limit, which means the characterized portion of our survey had a zero false-positive rate. This is the first time such a measurement of false-positive rate has been made, and we suggest that it should become standard. Our result of zero false positives in the characterized portion of the survey should not be seen as evidence that similar surveys always have a zero/low false-positive rate, but rather serves as evidence of the extensive effort invested in the candidate inspection process. If a significant number of false positives had appeared that were brighter than the characterization limit, it would call into question the validity of the efficiency function, and thus one's ability to de-bias the survey.

In the real triplets, there were 1246 candidates; 1159 of these were rejected, leaving 87 candidate real objects: 70 in the low-lat block and 17 in the high-lat block. To get the precise orbits needed for classification, these objects were then searched for through tracking observations in 2011–2013.

3.3. Tracking

We were able to track almost every object detected in our survey. Ten objects did not achieve two-opposition arcs, however, eight of these were past our characterization limit and thus do not affect our scientific modeling. Because the characterization indicated that we have a zero false-positive rate in our characterized survey, we believe that these are two real objects that were unsuccessfully tracked despite our efforts.

To account for the (slightly) less than 100% tracking efficiency in our analysis in the coming sections, we modeled the tracking efficiency based on the loss of these two objects, one in each block. We do not believe that our tracking efficiency is dependent on the orbit of the object, because our observations within the discovery year were always a large contiguous patch of sky, meaning tracking did not rely on assumed orbits. The low-lat untracked object, mal11nt = 2011 UU_{412} with $m_r = 24.5 \pm 0.1$, was near our characterization limit, so was simply invisible in the slightly shallower tracking images. We therefore modeled the tracking efficiency of the low-lat block as 100% for $m_r < 24.5$, followed by a linear falloff such that integrating over the tracking function gives the appropriate fraction of lost objects. In the high-lat block, mah11nt (no MPC designation due to short arc) with $m_r = 24.2 \pm 0.1$ was not successfully tracked, despite being a fair bit brighter than our limit. This was caused by the fact that mah11nt sheared off the field during part of the discovery year (including the nailing observations within the discovery month); its lack of recovery was thus independent of the object's brightness, so we modeled the tracking efficiency of the high-lat block as a constant 93%. These tracking efficiencies were incorporated into our survey simulations.

3.4. Orbit Classification of Real Objects

At the end of 2012B, our tracked objects typically had 17 month arcs, with 15–30 observations in multiple months in both 2011 and 2012. The cadence of observations, together with the superior astrometry, ensured that with just two oppositions the orbits of most of our objects were determined to far better precision than those of most two-opposition objects in the MPC database. To classify objects with reasonable confidence, relative semimajor axis uncertainties of $\sigma_a/a \lesssim 0.003$ are needed, and to constrain the libration amplitude of resonant objects, $\sigma_a/a \lesssim 0.001$ is typically required. Of our 77 characterized objects, 71 and 59 had σ_a/a smaller than these values, respectively, by the end of 2012B.

The objects were dynamically classified using the Solar System Beyond Neptune nomenclature (Gladman et al. 2008, p. 43). The nominal orbit, as well as the two orbits with the most extremal semimajor axis allowed by the astrometry, were integrated for 10 Myr. The two extremal orbits were determined from a Monte-Carlo process. If the integrated particles that started on these three orbits exhibit the same behavior (resonant, classical, detached, scattering, etc.), the object was said to be securely classified, but if all three clones did not experience similar behavior, the object was insecurely classified. Table 3 lists the resulting orbital classes and the security of the classification. Note that three objects were previously discovered objects that happened to be in our fields; however, all information in this work and in Table 3 (aside from the MPC designation) is derived solely from our observations.

In our discovery triplets, 87 objects were detected: 70 in the low-lat block and 17 in the high-lat block. Of these 87 objects, 77 were above our characterization limit, comprising our characterized sample. Among the characterized sample, we found one Uranian trojan (1:1 mean-motion resonance with Uranus; see Alexandersen et al. 2013a), two Neptunian trojans

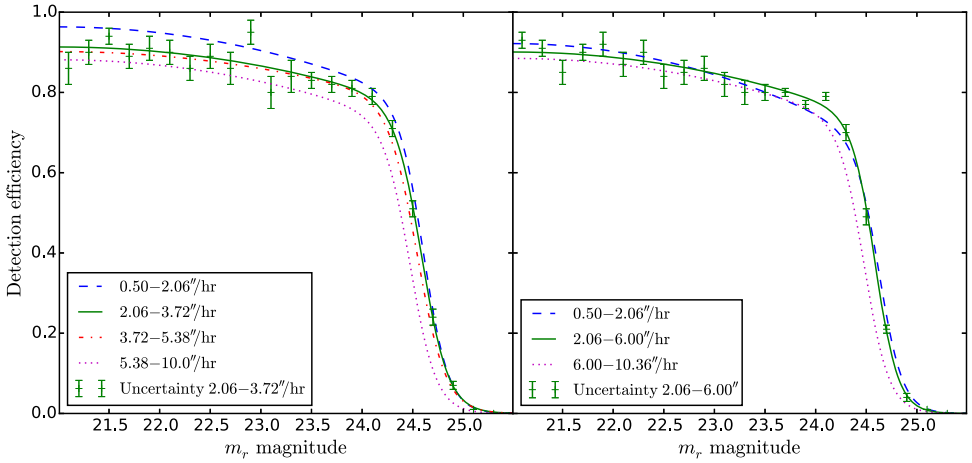


Figure 3. Efficiency functions for our survey low-lat block (left) and high-lat block (right). To prevent crowding, efficiency measurements and associated statistical error-bars are shown only on the curve most applicable for TNOs at ∼40 au.

(1:1 resonance with Neptune), two 4:3 resonant objects, 18 plutinos (3:2), six 5:3 objects, three 2:1 objects, two 5:2 objects, three 3:1 objects, one 4:1 object, and a slew of main belt and detached objects. We present the entire sample, with orbital information and classification, in Table 3. However, from here on we use the characterized sample exclusively, focusing on the plutinos in Section 4, Neptunian and Uranian co-orbitals in Section 5, and the 3:1 and 4:1 resonances in Section 6.

4. PLUTINO ANALYSIS

We analyzed the orbital and magnitude distribution of the plutinos (3:2 outer mean-motion resonance with Neptune) in detail. The plutinos are the most numerous population seen in our survey, aside from the classical belt. However, the classical belt is known to have a populous cold core that has a different absolute-magnitude distribution from the hot populations we are interested in. We therefore focused on the plutinos, aiming to investigate what constraints our plutino sample could put on the absolute H-magnitude distribution, orbital distribution, and population estimate of the plutinos.

Our plutino sample consists of 18 objects with osculating barycentric semimajor axis, eccentricity, inclination, and absolute-magnitude ranges of $39.21\,\mathrm{au} \leqslant a \leqslant 39.602\,\mathrm{au}$, $0.0436 \leqslant e \leqslant 0.3279$, $2.^\circ733 < i < 23.^\circ749$, and $5.58 \leqslant H_r \leqslant 9.85$. This plutino sample is thus, as planned, ideal for investigating the putative transition in the H distribution at $H_r = 7.7$–8.7. Details about these plutinos are presented in Table 3, including whether they undergo Lidov–Kozai oscillations, which is an added complication that we have not investigated in this work. All but one of these 18 objects have been found to be securely within the 3:2 resonance. The lowest eccentricity object, 2011 UB_{411}, is the only insecurely classified object, despite the eccentricity uncertainty of ± 0.0008 and semimajor axis uncertainty of 0.006 au. This insecurity is because the resonance is weaker and narrower at low eccentricity. However, because the region with the semimajor axis just larger than the 3:2 resonance is unstable and quickly cleared by the ν_8 secular resonance (Duncan et al. 1995), it is extremely likely that 2011 UB_{411} is a plutino.

In this work, we initially only used our sample of 18 plutinos, but because the CFEPS survey was characterized in a nearly identical fashion and detected 24 plutinos, we chose to include the characterization and detections from that survey to improve the statistics of our analysis. Most of our analysis is thus based on a sample of 42 plutinos with semimajor axis, eccentricity, inclination, and absolute-magnitude ranges of $39.21\,\mathrm{au} \leqslant a \leqslant 39.602\,\mathrm{au}$, $0.0436 \leqslant e \leqslant 0.3279$, $1.^\circ422 < i < 34.^\circ390$, and $5.58 \leqslant H \leqslant 9.85$. As six of the plutinos from our survey are fainter than any plutinos from CFEPS, our survey helps constrain the absolute-magnitude distribution for faint objects.

4.1. Statistical Method

Throughout this work, we used a modified version of the Anderson–Darling (AD) goodness of fit test (Anderson & Darling 1954) to test the hypothesis that a sample (either real observations or a small subset of simulated detections) could have been drawn from another, larger population (in our case always simulated). Because the calculated statistic (denoted $\mathcal{AD}$) is a normalized unitless quantity, it is possible to create a pseudo-multidimensional test. In the past, some (for example, the CFEPS papers) have created a multidimensional AD test by testing various distributions independently and using the highest rejectability for the rejectability of the model as a whole. However, this fails to take the "look elsewhere" effect into account, which means that the more distributions you look at, the more likely it will be that one of them is rejectable at 95% confidence; the CFEPS team knew of this shortcoming but were unable to correct for it (the "look elsewhere" effect can only be accurately accounted for when the distributions are independent, which is not the case for e, r and H). Others (e.g., Shankman et al. 2013), attempted to correct for this; instead of testing the rejectability of each distribution independently and selecting the highest one, they picked the worst statistic ($\mathcal{AD}_{\mathrm{max}} = \max(\mathcal{AD}_{\mathrm{dimension}})$) as representative and then bootstrapped this to find the rejectability of having such an $\mathcal{AD}_{\mathrm{max}}$. However, this can give a skewed view of the significance of the statistic, because a data set that gets a high statistic in all dimensions can have the same maximum statistic as a data set that only has a high statistic in one dimension, even though the one with multiple high statistics ought to get rejected more strongly. An alternate approach is to add the statistics from the various dimensions into a summed statistic (Parker 2015), $\sum \mathcal{AD} = \sum_{\mathrm{dimensions}} \mathcal{AD}_{\mathrm{dimension}}$. The sum is used because $\sum \mathcal{AD}$ will be small if the tested distributions are in

Table 3
List of all Objects Detected in Our Survey

ID	MPC	Disc	r (au)	N	t (year)	Epoch	a (au)	e	i (°)	Ω (°)	ω (°)	M (°)	m_r	H_r	S	C_{pq} (°)	L_{pq} (°)	C_{LK} (°)	L_{LK} (°)
Temporarily resonant 1:1 with Uranus (Uranian co-orbital) (Alexandersen et al. 2013b)																			
mal01	2011 QF_{99}	5858.9	20.296(0)	33	1.4722	5803.0	19.092(1)	0.17687(9)	10.810(0)	222.499(0)	287.47(2)	258.64(2)	22.6(1)	9.57	S	$\sim 70^{\sim 70}_{\sim 70}$	$\sim 30^{\sim 30}_{\sim 30}$	...	...
Stable resonant 1:1 (Neptunian trojan) (Alexandersen et al. 2014f)																			
mah02	2012 UV_{177}	6220.9	29.578(1)	30	1.4614	5805.9	30.024(4)	0.0723(9)	20.833(0)	265.668(2)	204.28(10)	285.94(2)	23.7(1)	8.93	S	60^{60}_{61}	12^{13}_{18}	...	...
Temporarily resonant 1:1 (Neptunian co-orbital) (Tomatic 2014a)																			
mah01	2012 UW_{177}	6220.9	22.432(1)	17	1.0429	6158.0	30.072(3)	0.2591(2)	53.886(1)	20.010(1)	34.4(2)	351.967(4)	24.2(1)	10.65	S	$\sim 60^{\sim 60}_{\sim 60}$	$\sim 25^{\sim 25}_{\sim 25}$	...	...
Resonant 4:3 (Alexandersen et al. 2014m)																			
mal09	2011 UZ_{412}	5858.9	31.606(1)	24	1.4422	5803.0	36.433(3)	0.13295(12)	6.812(1)	9.788(3)	27.4(7)	355.932(1)	24.0(1)	9.06	S	...	17^{21}_{14}	...	...
mal10	2011 UA_{413}	5860.8	31.794(3)	19	1.2866	5859.8	36.528(9)	0.1329(10)	5.437(0)	292.353(13)	83(2)	11.076(5)	23.3(1)	8.29	S	...	68^{79}_{64}	...	...
Resonant 3:2 (plutino) (Alexandersen et al. 2014a; Tomatic 2014b)																			
mal45	2011 UD_{411}	5860.8	41.480(1)	28	2.2668	5804.0	39.21(3)	0.2884(11)	15.700(1)	19.081(1)	131.88(11)	274.68(9)	23.1(1)	6.91	S	...	$130^{105}_{\sim 150}$	...	...
mal04	2011 UA_{411}	5860.8	29.828(1)	29	2.1003	5804.0	39.223(4)	0.24459(9)	8.645(0)	237.962(2)	167.51(9)	351.158(1)	23.7(1)	9.01	S	...	123^{105}_{123}	...	...
mal02	2011 UC_{411}	5860.8	28.724(0)	43	2.0894	5804.0	39.270(3)	0.27179(9)	4.652(0)	289.175(5)	88.35(6)	6.447(1)	22.3(1)	7.74	S	...	92^{86}_{98}	...	...
mal03	2011 UU_{410}	5858.9	28.997(1)	19	2.0893	5803.0	39.293(4)	0.26294(7)	12.623(1)	19.330(1)	15.39(8)	356.454(1)	24.4(1)	9.85	S	...	94^{90}_{97}	...	...
mah03	2012 UG_{177}	6220.9	30.482(2)	18	1.2344	5889.9	39.304(10)	0.2248(3)	21.876(1)	261.872(2)	130.6(7)	357.684(1)	24.4(1)	9.48	S	...	86^{77}_{92}	90	$42.4^{43.4}_{43.9}$
mal30	2004 VT_{75}	5858.9	37.958(1)	31	2.2717	5803.0	39.31(2)	0.2108(10)	12.842(1)	26.610(0)	272.04(3)	68.66(5)	21.9(1)	6.11	S	...	70^{100}_{45}	270	55^{57}_{51}
mal16	2011 UY_{410}	5860.8	34.137(2)	24	1.4313	5804.0	39.312(8)	0.1364(7)	8.590(1)	12.404(2)	2.9(8)	13.169(5)	24.4(1)	9.09	S	...	79^{74}_{82}	...	...
mah15	2012 UH_{177}	6221.9	45.464(2)	27	1.4559	5806.0	39.39(2)	0.1907(12)	23.749(1)	255.149(3)	346.2(6)	137.39(10)	24.2(2)	7.57	S	...	41^{77}_{16}	...	...
mal07	2011 UQ_{410}	5858.9	30.974(1)	26	2.0071	5803.0	39.400(8)	0.2636(3)	3.246(1)	10.821(4)	335.40(6)	26.946(8)	24.2(1)	9.36	S	...	69^{81}_{73}	...	...
mal21	2011 UX_{410}	5860.8	36.258(1)	28	2.1738	5804.0	39.40(2)	0.3279(7)	17.479(1)	223.962(0)	263.515(11)	302.35(4)	23.7(1)	8.14	S	...	63^{134}_{65}	270	$85^{!}_{83}$
mal06	2011 UR_{410}	5858.9	30.875(1)	21	2.2693	5803.0	39.412(9)	0.2685(4)	17.320(1)	27.050(0)	50.37(9)	332.881(9)	24.4(1)	9.59	S	...	53^{66}_{62}	90	40^{40}_{40}
mal05	2011 UV_{410}	5858.9	29.862(1)	26	2.1766	5803.0	39.421(7)	0.2591(3)	4.308(0)	332.153(6)	30.14(9)	15.396(4)	22.5(1)	7.77	S	...	73^{77}_{72}	...	...
mal17	2011 US_{410}	5858.9	35.150(1)	27	2.1028	5803.0	39.510(13)	0.2554(5)	11.876(1)	20.578(1)	290.41(2)	51.18(3)	23.7(1)	8.27	S	...	97^{128}_{76}	270	$47.2^{!}_{47.2}$
mal35	2011 UT_{410}	5858.9	39.355(1)	24	2.2420	5803.0	39.52(2)	0.1787(11)	13.498(1)	19.796(1)	109.09(4)	281.59(5)	21.5(1)	5.58	S	...	$114^{\sim 150}_{108}$	...	...
mal31	2011 UZ_{410}	5860.8	38.402(1)	26	2.3184	5804.0	39.521(13)	0.1245(12)	3.833(0)	298.461(10)	8.44(4)	69.87(4)	24.0(1)	8.17	S	...	79^{108}_{47}	...	...
mal28	2011 UW_{410}	5860.8	37.600(1)	30	2.1631	5804.0	39.578(14)	0.2175(7)	10.752(1)	17.914(1)	285.455(14)	64.55(3)	23.8(1)	8.07	S	...	120^{143}_{71}	...	...
mal32	2011 UB_{411}	5860.8	38.548(1)	25	2.0862	5804.0	39.601(6)	0.0436(8)	14.900(1)	13.374(1)	322.5(6)	50.471(14)	23.8(1)	7.92	I	...	$\sim 130^{\sim 150}_{\sim 150}$	...	...
mal12	2011 UP_{410}	5858.9	33.293(1)	32	2.2503	5810.0	39.602(4)	0.1749(3)	2.733(1)	241.740(6)	179.2(2)	339.745(4)	23.4(1)	8.21	S	...	113^{127}_{100}	...	...
Resonant 5:3 (Alexandersen et al. 2014h)																			
mal60	2011 UM_{411}	5858.9	48.408(7)	12	1.2891	5858.9	42.222(14)	0.1467(4)	9.385(3)	11.445(6)	197(4)	183.8(2)	24.1(1)	7.24	S	...	41^{41}_{51}	...	...
mal34	2011 UL_{411}	5858.9	39.189(2)	16	1.4340	5803.0	42.224(5)	0.07190(9)	1.974(0)	308.53(4)	82(2)	358.747(0)	24.3(1)	8.37	S	...	87^{68}_{98}	...	...
mal14	2011 UO_{411}	5860.8	33.891(1)	26	1.4395	5804.0	42.282(11)	0.2061(6)	5.289(0)	310.100(10)	98.9(5)	347.583(5)	23.8(1)	8.51	S	...	18^{33}_{15}	...	...
mal27	2011 UJ_{411}	5858.9	37.500(2)	21	1.4338	5803.0	42.290(9)	0.1189(7)	5.141(1)	21.604(3)	30.0(1.0)	344.336(6)	23.9(1)	8.20	S	...	31^{51}_{23}	...	...
mal15	2011 UK_{411}	5858.9	33.939(1)	27	1.4724	5803.0	42.32(2)	0.2328(11)	13.424(1)	219.278(1)	210.3(3)	335.30(2)	21.9(1)	6.66	S	...	41^{77}_{16}	...	...
mal13	2011 UN_{411}	5860.8	33.447(2)	23	1.4395	5804.0	42.341(7)	0.2104(3)	3.234(0)	326.45(2)	61.0(7)	2.553(1)	24.2(2)	8.99	S	...	55^{68}_{46}	...	...
Resonant 19:10 (Alexandersen et al. 2014c)																			
mal18	2011 UH_{411}	5858.9	35.636(2)	18	1.4504	5803.0	46.14(2)	0.2386(9)	27.041(1)	214.057(0)	198.3(5)	346.696(9)	24.2(1)	8.72	I	...	$\sim 150^{!}_{\sim 150}$	...	...
Resonant 2:1 (Alexandersen et al. 2014j)																			
mal22	2011 UE_{411}	5858.9	36.574(2)	14	1.3792	5803.1	47.61(9)	0.320(3)	9.355(1)	17.099(3)	73.3(3)	329.11(9)	24.6(1)	9.00	S	76^{73}_{180}	24^{23}_{164}	...	...

Table 3
(Continued)

ID	MPC	Disc	r (au)	N	t (year)	Epoch	a (au)	e	i (°)	Ω (°)	ω (°)	M (°)	m_r	H_r	S	C_{pq} (°)	L_{pq} (°)	C_{LK} (°)	L_{LK} (°)
mal47	2011 UG_{411}	5860.8	41.990(4)	18	1.2784	5859.8	47.66(4)	0.139(3)	10.692(1)	238.833(4)	116(2)	27.11(4)	24.6(2)	8.35	S	97^{89}_{180}	42^{21}_{159}	…	…
mal25	2011 UF_{411}	5860.8	37.267(2)	31	1.4313	5804.0	47.73(9)	0.344(2)	5.872(1)	261.775(10)	195.8(2)	324.76(10)	24.1(1)	8.35	S	70^{91}_{90}	14^{52}_{51}	…	…
Resonant 5:2 (Alexandersen et al. 2014e)																			
mah04	2012 UJ_{177}	6221.9	34.429(1)	25	1.4585	5805.0	55.20(7)	0.4333(13)	15.632(0)	298.867(3)	46.9(2)	17.41(4)	24.1(1)	8.65	I	…	$104^{71}_{\sim 150}$	…	…
mal61	2011 UT_{411}	5860.8	51.239(4)	20	1.4338	5803.1	55.7(3)	0.406(5)	6.420(2)	4.876(11)	130.2(2)	304.2(4)	24.5(1)	7.46	I	…	$158^{!}_{164}$	…	…
Resonant 3:1 (Alexandersen et al. 2014k)																			
mal23	2011 UR_{411}	5860.8	37.046(3)	15	1.2866	5859.9	62.1(2)	0.438 (3)	26.581(2)	23.363(1)	45.8(6)	346.99(7)	24.0(1)	8.33	I	$180^{72}_{!}$	$\sim 170^{20}_{!}$	…	…
mal08	2011 US_{411}	5860.8	31.226(2)	21	1.4308	5810.1	62.431(11)	0.49983(8)	22.040(1)	24.813(0)	9.4(3)	359.981(0)	24.0(1)	9.11	S	69^{67}_{71}	19^{14}_{25}	…	…
mal62	2011 UQ_{411}	5860.8	52.005(2)	25	1.4396	5804.0	62.4(2)	0.405 (4)	40.400(2)	215.504(0)	263.65(2)	315.5(2)	23.9(1)	6.76	I	$180^{!}_{!}$	$\sim 160^{!}_{!}$	…	…
Resonant 4:1 (Alexandersen et al. 2014d)																			
mal33	2011 UP_{411}	5860.8	38.569(3)	17	1.4228	5810.1	75.79(2)	0.49117(15)	13.435(1)	231.961(2)	160.3(5)	359.450(0)	24.2(1)	8.35	S	77^{78}_{76}	19^{23}_{18}	…	…
Resonant 16:3 (Alexandersen et al. 2014l)																			
mah08	2012 UK_{177}	6220.9	36.784(1)	25	1.4560	5805.9	92.3(2)	0.6173(9)	24.852(1)	251.482(2)	162.5(2)	354.981(14)	24.1(1)	8.38	I	…	$\sim 150^{!}_{\sim 110}$	…	…
Inner classical (Alexandersen et al. 2014n)																			
mal29	2011 UO_{412}	5860.8	37.892(2)	26	1.4146	5810.1	38.04(2)	0.129(3)	27.787(1)	21.128(0)	273.13(12)	80.85(8)	23.4(1)	7.62	I	…	…	…	…
mal41	2011 UN_{412}	5858.9	40.960(4)	13	1.2892	5858.9	38.96(2)	0.058(3)	20.411(2)	216.340(1)	326(5)	209.1(2)	24.5(1)	8.37	S	…	…	…	…
Main classical (Alexandersen et al. 2014g; Tomatic 2014b)																			
mah05	2012 UL_{177}	6220.9	36.474(1)	30	1.4643	5805.9	40.666(9)	0.1166(8)	19.236(0)	272.168(3)	147.6(0.6)	335.255(9)	22.9(1)	7.24	S	…	…	…	…
mal24	2011 UK_{412}	5860.8	37.131(2)	29	1.4476	5804.0	40.737(7)	0.0920(7)	26.365(1)	218.923(1)	151.6(1.3)	14.430(5)	23.4(1)	7.74	S	…	…	…	…
mah13	2012 UO_{177}	6221.9	42.380(2)	25	1.4560	5805.9	41.562(5)	0.035(2)	16.446(0)	289.959(6)	336(2)	122.36(6)	24.4(1)	8.13	S	…	…	…	…
mal50	2011 UJ_{412}	5860.8	42.997(2)	17	1.4338	5803.1	42.031(7)	0.029(2)	29.757(2)	27.031(0)	148(5)	219.82(10)	24.3(2)	7.99	S	…	…	…	…
mal37	2011 UX_{411}	5858.9	39.653(2)	23	1.3793	5803.1	42.43(2)	0.098(2)	7.210(1)	24.163(2)	315.7(1.0)	44.02(3)	24.2(2)	8.27	I	…	…	…	…
mal44	2011 UH_{412}	5860.8	41.372(2)	22	1.4313	5804.0	42.60(3)	0.116(3)	15.118(2)	21.517(1)	94.4(2)	290.88(7)	24.2(1)	8.02	S	…	…	…	…
mah09	2012 UP_{177}	6221.9	39.922(2)	27	1.4585	5805.0	42.823(14)	0.099(2)	16.601(1)	280.170(4)	59.9(8)	42.49(2)	24.0(1)	7.97	S	…	…	…	…
mal49	2011 UL_{412}	5860.8	42.833(2)	21	1.4395	5804.0	43.03(2)	0.083(3)	10.371(2)	239.455(5)	60.14(10)	82.06(7)	23.7(1)	7.42	I	…	…	…	…
mal40	1999 RU_{195}	5858.9	40.875(2)	24	1.4339	5803.0	43.048(12)	0.071(2)	7.737(1)	14.293(3)	329.4(1.4)	41.43(2)	23.1(1)	7.04	S	…	…	…	…
mah07	2012 UN_{177}	6220.9	36.611(2)	29	1.4667	5805.0	43.181(10)	0.1589(6)	20.094(1)	341.291(2)	71.2(7)	345.867(6)	22.9(1)	7.28	S	…	…	…	…
mal43	2011 UV_{411}	5858.9	41.355(2)	23	1.4340	5803.0	43.363(13)	0.067(2)	4.419(1)	20.174(4)	323(2)	43.37(3)	24.1(1)	7.93	S	…	…	…	…
mal39	2011 UC_{412}	5858.9	40.517(2)	26	1.4723	5803.0	43.408(6)	0.0705(8)	3.132(1)	337.47(2)	33(2)	17.794(7)	23.8(1)	7.75	S	…	…	…	…
mal59	2011 UE_{412}	5858.9	48.245(3)	26	1.4339	5803.0	43.496(5)	0.1119(5)	22.799(2)	218.117(0)	5.6(1.4)	165.93(6)	23.3(1)	6.45	S	…	…	…	…
mal54	2011 UB_{412}	5858.9	44.937(5)	17	1.2809	5858.9	43.50(2)	0.056(5)	15.311(2)	14.512(2)	246(4)	123.94(14)	24.3(1)	7.76	S	…	…	…	…
mah12	2012 UM_{177}	6220.9	41.687(2)	19	1.4586	5805.0	43.549(11)	0.058(2)	16.177(0)	287.738(6)	57(2)	40.52(2)	24.6(2)	8.33	S	…	…	…	…
mal46	2011 UW_{411}	5858.9	41.526(2)	20	1.4340	5803.0	43.84(4)	0.165(3)	4.529(2)	18.484(5)	94.7(2)	297.62(8)	24.5(2)	8.37	I	…	…	…	…
mal52	2011 UU_{411}	5858.9	43.301(2)	24	1.4723	5803.0	43.941(8)	0.031(2)	1.694(1)	357.00(3)	99(2)	299.72(3)	24.3(1)	7.96	S	…	…	…	…
mal42	2001 RZ_{143}	5858.9	41.288(2)	23	1.3795	5803.0	44.061(5)	0.0640(5)	2.122(2)	8.32(2)	35(3)	350.411(4)	22.6(1)	6.44	S	…	…	…	…
mal51	2011 UY_{411}	5858.9	43.191(2)	25	1.4423	5803.0	44.22(2)	0.064(2)	1.907(0)	309.43(4)	154.1(7)	294.69(4)	24.0(1)	7.64	S	…	…	…	…
mal56	2011 UF_{412}	5858.9	45.576(4)	12	1.1466	5803.1	45.07(2)	0.012(3)	2.146(0)	313.97(7)	280(40)	160.7(9)	24.4(1)	7.83	S	…	…	…	…
mal38	2011 UM_{412}	5860.8	40.339(2)	27	1.4309	5804.1	45.38(2)	0.132(2)	10.729(1)	231.692(2)	124.1(1.0)	28.50(2)	24.4(2)	8.36	I	…	…	…	…
mal53	2011 UD_{412}	5858.9	44.398(1)	22	2.0893	5803.0	45.419(6)	0.0535(11)	5.496(1)	358.349(7)	101.0(5)	297.64(1)	23.2(1)	6.75	S	…	…	…	…
mal55	2011 UZ_{411}	5858.9	45.158(2)	22	1.4504	5803.0	45.535(14)	0.053(3)	1.214(0)	315.36(7)	352.2(3)	77.97(4)	24.3(1)	7.75	S	…	…	…	…
mal36	2011 UA_{412}	5858.9	39.507(3)	12	1.1468	5803.0	46.250(11)	0.1476(8)	2.956(1)	252.56(2)	146(2)	352.267(4)	24.6(2)	8.66	I	…	…	…	…
mal48	2011 UG_{412}	5860.8	42.436(2)	33	1.4313	5804.0	46.45(3)	0.155(2)	2.496(0)	303.96(3)	24.9(4)	48.71(6)	23.8(1)	7.57	S	…	…	…	…

Table 3
(Continued)

ID	MPC	Disc	r (au)	N	t (year)	Epoch	a (au)	e	i (°)	Ω (°)	ω (°)	M (°)	m_r	H_r	S	C_{pq} (°)	L_{pq} (°)	C_{LK} (°)	L_{LK} (°)
Outer classical (Alexandersen et al. 2014o)																			
mal58	2011 US_{412}	5858.9	46.661(4)	16	1.4501	5803.1	47.88(8)	0.161(5)	2.606(2)	5.09(2)	116.38(4)	288.2(2)	24.4(1)	7.74	S	…	…	…	…
mal57	2011 UT_{412}	5860.8	46.124(3)	15	1.4395	5804.0	48.08(7)	0.183(4)	17.822(2)	224.314(2)	79.71(5)	66.9(2)	24.1(1)	7.50	I	…	…	…	…
Detached (Alexandersen et al. 2014i)																			
mal20	2011 UR_{412}	5860.8	36.158(2)	20	1.4395	5804.0	50.07(6)	0.314(2)	17.873(1)	15.280(1)	54.2(4)	340.63(3)	24.2(2)	8.60	S	…	…	…	…
mah10	2012 UQ_{177}	6220.9	40.257(2)	26	1.4560	5805.9	51.97(7)	0.312(2)	19.603(1)	330.160(3)	120.1(2)	328.58(7)	23.8(1)	7.77	S	…	…	…	…
mah06	2012 US_{177}	6221.9	36.605(2)	25	1.4665	5805.1	56.09(5)	0.3665(11)	17.214(1)	271.782(3)	92.5(3)	11.89(2)	24.2(1)	8.51	S	…	…	…	…
mal26	2011 UP_{412}	5858.9	37.450(2)	23	1.4422	5803.0	56.34(3)	0.3443(7)	19.721(1)	23.081(0)	26.4(4)	351.390(8)	23.8(1)	8.09	S	…	…	…	…
mal19	2011 UQ_{412}	5858.9	35.912(1)	22	1.4311	5803.0	67.32(8)	0.4829(9)	16.712(1)	219.944(0)	197.5(2)	352.205(14)	23.0(1)	7.51	S	…	…	…	…
mah14	2012 UR_{177}	6221.9	44.388(3)	26	1.4560	5805.9	73.8(3)	0.492(3)	16.353(0)	291.500(6)	158.2(2)	340.65(13)	24.2(1)	7.76	I	…	…	…	…
Unclassified ("lost") (Tomatic 2014b)																			
mah11nt		6221.9	38(4)	3	0.0002	6221.9	39(20)	0.0(6)	15(4)	320(80)	90(1400)	352(7)	24.2(1)	8.30	N	…	…	…	…
mal11nt	2011 UU_{412}	5860.8	32.8(5)	11	0.0986	5859.8	39(10)	0.2(5)	23(2)	25.5(7)	30(250)	343(7)	24.5(1)	9.40	N	…	…	…	…
Uncharacterized & unclassified (Alexandersen et al. 2014b; Tomatic 2014b)																			
umal64nt		5858.9	36(4)	3	0.0002	5858.9	40(20)	0.0(6)	3(3)	300(300)	100(1300)	0.8(7)	24.7(3)	9.09	N	…	…	…	…
umal70nt	2011 UV_{412}	5858.9	47.85(5)	9	0.8354	5858.9	48(5)	0.3(2)	11.54(3)	219.29(3)	63(11)	74(12)	24.7(1)	7.89	N	…	…	…	…
umal65nt	2011 UY_{412}	5858.9	42(2)	4	0.0195	5858.9	40(20)	0.0(6)	8(3)	16(6)	20(1500)	357(3)	24.7(1)	8.52	N	…	…	…	…
umah16nt	2012 UU_{177}	6221.9	45(2)	7	0.0055	6220.0	50(20)	0.0(6)	16.1(6)	284(5)	120(1800)	352(8)	24.9(3)	8.30	N	…	…	…	…
umal66nt		5858.9	46(5)	3	0.0002	5858.9	50(20)	0.0(6)	13(25)	20(20)	10(1600)	357(3)	24.6(1)	8.06	N	…	…	…	…
umah17	2012 UT_{177}	6221.9	52.65(1)	21	1.4614	5806.0	47.49(8)	0.127(7)	16.363(1)	320.497(13)	282(6)	144.9(5)	24.8(1)	7.49	N	…	…	…	…
umal68nt		5858.9	47(5)	3	0.0002	5858.9	50(30)	0.0(6)	8(24)	10(60)	20(1700)	0.7(7)	24.8(1)	8.04	N	…	…	…	…
umal69nt		5858.9	47(5)	3	0.0002	5858.9	50(30)	0.0(6)	3(20)	355(304)	40(1700)	358(2)	24.8(1)	8.10	N	…	…	…	…
umal63	2011 UW_{412}	5860.8	38.419(3)	18	1.4228	5810.1	78.2(2)	0.521(2)	12.976(1)	233.178(2)	134.1(5)	6.17(3)	24.7(2)	8.86	N	…	…	…	…
umal67nt	2011 UX_{412}	5858.9	46.9(4)	7	0.1010	5858.9	50(30)	0.4(4)	29(2)	216.9(4)	270(16)	310(30)	24.6(2)	7.94	N	…	…	…	…

Note. ID is the internal designation, "ma" denotes the survey, and "*l*"/"*h*" indicates the *l*ow-lat or *h*igh-lat block. MPC is the minor planet center designation. The date of discovery (given as the Julian Date–2,450,000) is Disc and the heliocentric distance at discovery is r. N and t are the total number of astrometric measurements and total arc in years, respectively. The J2000 barycentric orbital elements, semimajor axis (a), eccentricity (e), inclination (i), longitude of ascending node (Ω), argument of pericenter (ω), and mean anomaly (M) are all given at the Epoch (Julian date−2,450,000). The average r-band apparent magnitude, measured by the moving-object pipeline, in the discovery triplet is m_r. H_r is the absolute magnitude calculated from m_r using the appropriate phase-angle corrections. Uncertainties on the last digit(s) of r, a, e, i, Ω, ω, M, and m_r are in parentheses after the respective value; the uncertainty for the orbital elements is from the Bernstein & Khushalani (2000) orbit-fitting code. Column S denotes whether an object is securely (S), insecurely (I) or not (N) classified. The libration center C_{pq} is given for resonances that allow values other than 180°, while the libration amplitude L_{pq} is given for all resonant objects. Objects that are additionally in the Lidov–Kozai resonance (Kozai 1962; Lidov 1962) also have the libration center C_{LK} and libration amplitude L_{LK} listed. C_{pq}, L_{pq}, and L_{LK} have values listed for the nominal fit, the Max-a clone (superscript) and Min-a clone (subscript) used during classification, because we do not have a good way of estimating the uncertainty on these. A clone that was not in the resonance has a !, while a ∼ before a value means the value had to be manually estimated. The minor planet electronic circulars containing our astrometry are cited next to each class header.

(This table is available in its entirety in machine-readable form.)

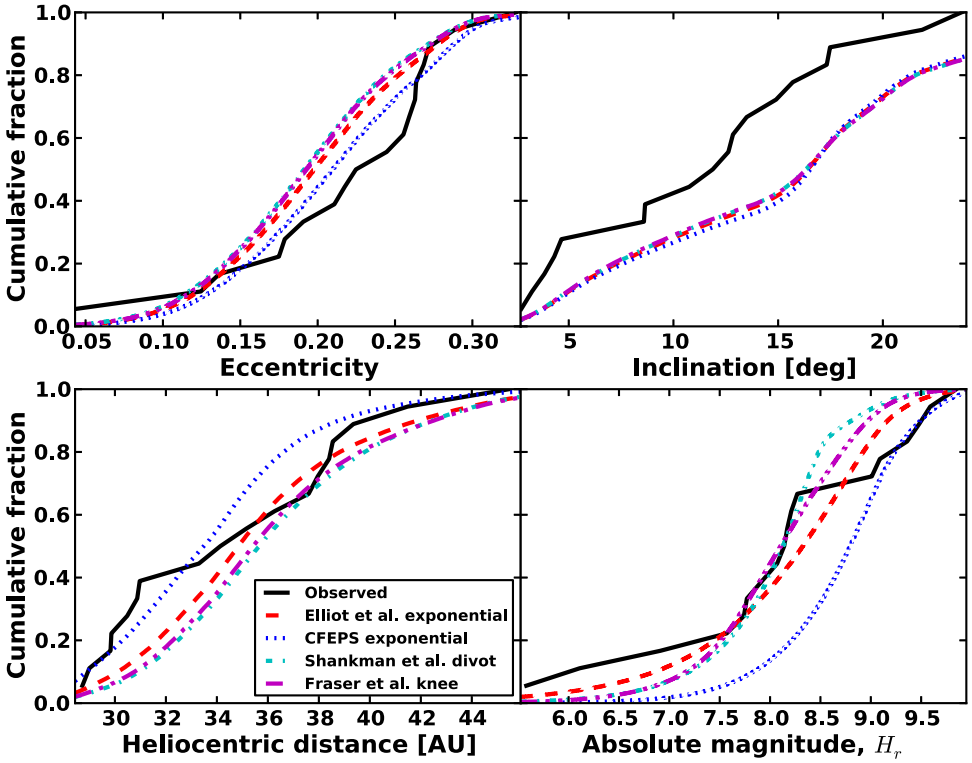


Figure 4. Clockwise from top left: cumulative distributions of the eccentricity, inclination, absolute magnitude at discovery, and heliocentric distance at discovery for the real objects (solid black lines) detected in our survey, spanning the full (uncut) H-magnitude range. Simulated detections from the magnitude distributions in Elliot et al. (2005), Gladman et al. (2012), Shankman et al. (2013), and Fraser et al. (2014) are plotted for comparison. Notice that the inclination distribution of the real detections differs significantly from the simulated ones in all cases, suggesting that the underlying intrinsic inclination distribution is not as wide as that used in the model. The reader is cautioned that the intrinsic distribution is not shown, only the biased detections after survey simulation.

good agreement in all dimensions (all $\mathcal{AD}_{\text{dimension}}$ are low), large if one of the dimensions has a poor match, and even larger if several dimensions show large discrepancies between the data and model (Parker 2015). Note that the summed statistic is thus not quite as sensitive to rejecting cases where a single distribution gives a large discrepancy, because the same summed statistic could be caused by several dimensions being moderately discrepant. Inversely, the summed statistic will be slightly stronger at rejecting cases where multiple dimensions are moderately discrepant than the maximum statistic method was. Ideally, a model should represent a data set well in all dimensions, and we therefore choose to use the summed statistic because it rejects models that are discrepant with the data in a broader sense.

To determine the rejectability of a model given its $\sum\mathcal{AD}$ statistic, the distribution of $\sum\mathcal{AD}$ for samples (equal in size to the real sample) drawn from the comparison model has to be calculated. This was done by calculating $\sum\mathcal{AD}_{\text{sim}}$ for 10^3–10^4 simulated data sets of the same size as the real data set (usually 12–42), drawn from the comparison model, which was usually a large sample of 1×10^4–5×10^4 simulated detections generated with our survey simulator (Petit et al. 2011). The summed AD statistic for our real observations, $\sum\mathcal{AD}_{\text{real}}$ is then calculated and compared with the distribution of $\sum\mathcal{AD}_{\text{sim}}$. The distribution of summed AD statistics for the simulated subsets reveals the probability of a random subset being a worse match to the parent simulated detections than the real observations are, giving us the significance of the statistic of the real sample. The rejectability of the hypothesis that the real detections were drawn from the model is

$$P(\mathcal{AD}_{\text{real}}) = \frac{N_{\text{sim}\,>\,\text{real}}}{N_{\text{sim}}}$$

where $N_{\text{sim}>\text{real}}$ is the number of times $\mathcal{AD}_{\text{sim}} > \mathcal{AD}_{\text{real}}$, and N_{sim} is the total number of simulated $\mathcal{AD}_{\text{sim}}$. This is analogous to how Kolmogorov-Smirnov (KS) tests are usually interpreted. In this work, probabilities $<5\%$ (rejection at $>95\%$ confidence) or $<1\%$ (rejection at $>99\%$ confidence) are considered thresholds of significant or highly significant rejection, respectively.

For our analysis of the plutinos in this work, we used a four-dimensional AD test as described above, using four properties: eccentricity, inclination, heliocentric distance at discovery, and absolute magnitude at discovery. These parameters were chosen because they are the most important parameters in revealing the significant shortcomings of a combined orbit and magnitude distribution model; for example, because all plutinos have semimajor axis values within 1% of each other, detailed modeling of the a distribution is unimportant, as small semimajor axis variations in this range have no discernible effect on detectability. However, changing the eccentricity model or absolute-magnitude model within reason changes the eccentricity, distance, and absolute-magnitude distributions of detections making them highly important for model testing. The inclination model is less coupled than the e, H, and r distributions, so it could be tested separately; either way, it is important to constrain the inclination distribution, because changing the inclination model changes the population estimate.

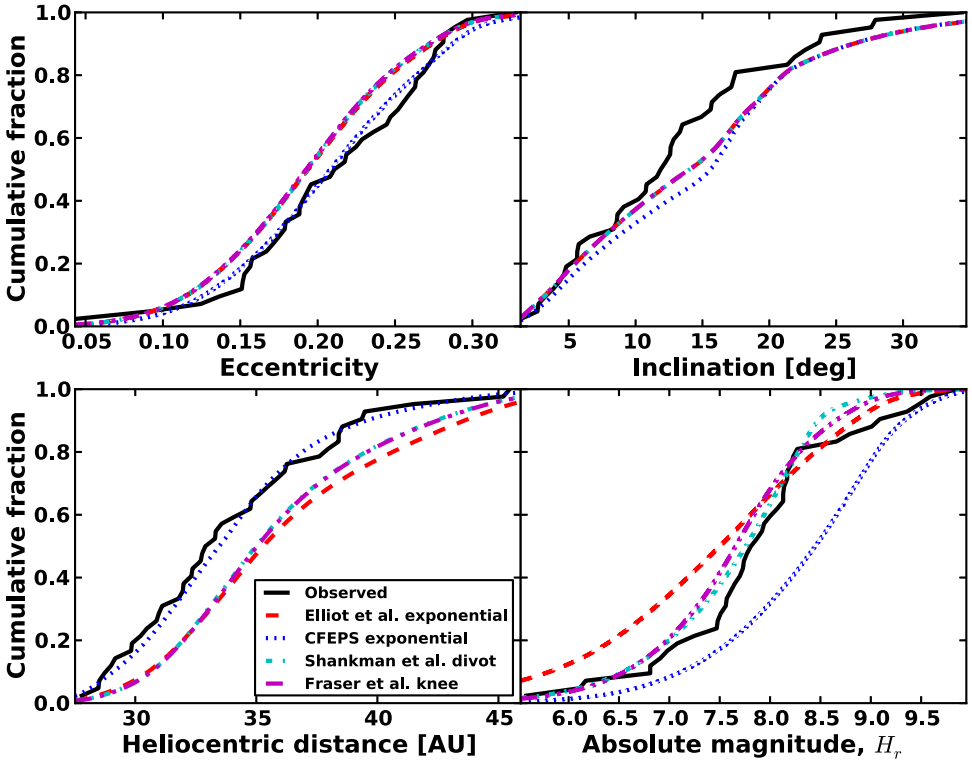


Figure 5. Cumulative distributions for the combined $H_r < 11.0$ sample, similar to those in Figure 4, but for the combined surveys (our new survey combined with CFEPS) with the full range of H-magnitudes ($H_r < 11$). Plotted for comparison are simulated detections from the magnitude distributions in Elliot et al. (2005), Gladman et al. (2012), Shankman et al. (2013), and Fraser et al. (2014), extrapolated to $H_r = 11$; these models are rejectable at 99.8%, 98.7%, 99.2%, and 99.4% confidence, respectively. Note how the extrapolated CFEPS model appears to match the eccentricity and heliocentric distance, but is rejectable due to the absolute magnitude, while the knee and divot have fairly good absolute-magnitude distributions, but are rejectable due to the eccentricity and heliocentric distance. All models predict more high-inclination objects than observed, suggesting that some of the models might be made non-rejectable by adjusting the inclination model.

4.2. Refinements to the CFEPS L7 Model and Survey Simulator

Our detections required two minor changes to the previous orbital model: the CFEPS L7 model (Gladman et al. 2012). Three of the plutinos detected in our survey have a combination of a and e outside the model resonance boundary used by CFEPS. Our new resonance bounds were enlarged to ensure that all the real observed plutinos are within the bounds; because our numerical integrations had shown these object to be resonant, they must be within the real bounds of the resonance. Because all plutinos have semimajor axes within ∼1% of each other, the slight change in semimajor axis range does not have a noticeable influence on detectability. Having widened the low-eccentricity ($e < 0.16$) part of the resonance, where the resonance narrows there is now slightly more phase space with $e < 0.16$.

The second change to the the model was to correct an implementation error in the libration-amplitude distribution. Rather than having a triangular distribution from 20° to 130° peaking at $L_{\rm peak}$, as described in Gladman et al. (2012), the old model object generator had a triangle from 0° for which values of <20° were truncated. We also changed the maximum end of the triangle to 135°, because two of our plutinos have libration amplitudes of ∼130°.

These combined changes led to slightly fewer low-amplitude objects and slightly more low-eccentricity objects being generated; however, this only increased the population estimate by ∼4%. This is insignificant in comparison with the effects of recent updates to the survey simulator itself; since Gladman et al. (2012), the survey simulator has had major updates, among which was a significant improvement to how magnitude uncertainty is handled. We found that in some situations, these updates led to changes of 10%–20% in the population estimate. The effects of the improvements in the survey simulator thus vastly outweigh the effects of our minor improvements in the a–e and libration-amplitude models. Henceforth, the terms L7 model and survey simulator refer to the updated versions described in this section.

4.3. Testing Previously Published Models

The magnitude limits of most of the CFEPS sky blocks were more than a magnitude brighter than those for our observations, but CFEPS imaged a much larger area, resulting in 24 plutino detections. The CFEPS L7 model of the plutino H-magnitude and orbital distributions was constructed based on these 24 plutinos with $6.1 < H_r < 8.8$ (Gladman et al. 2012). As in this work, the primary motivation was to obtain a population estimate, rather than measure details of the orbital distribution. However, to calculate a population estimate we must ensure that we use a model that is not rejectable by the data.

4.3.1. Comparison of Our 18 Plutinos with Published Models

We first investigated whether our independent plutino sample rejected the CFEPS L7 model if the latter's H magnitude distribution was extrapolated beyond its defined range. This was done by running survey simulations of just our

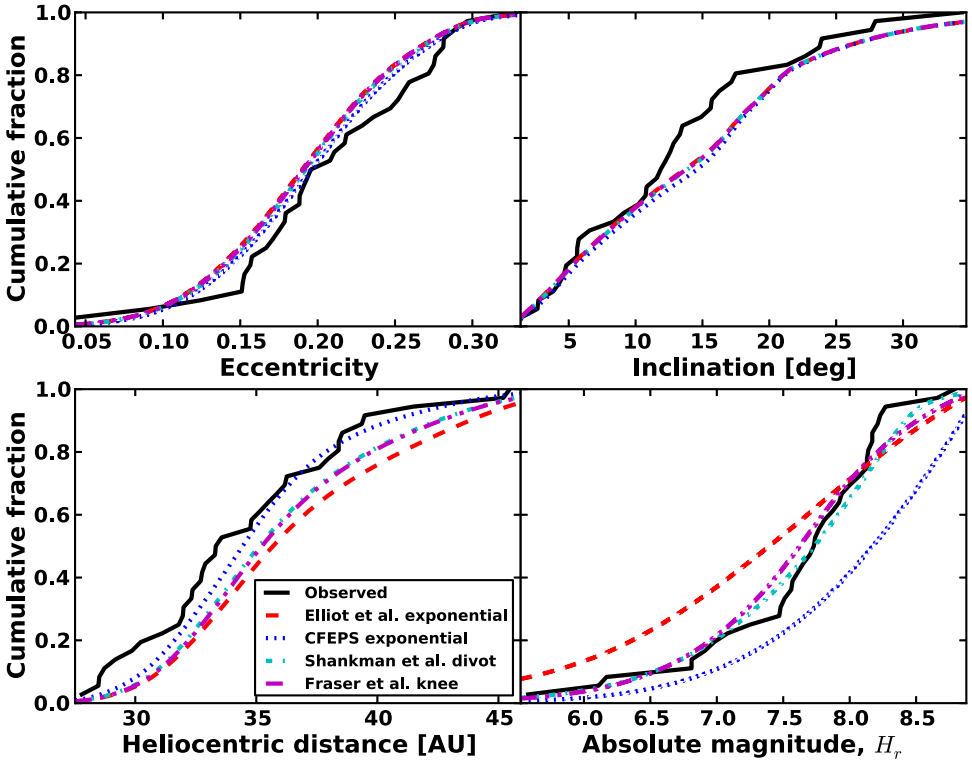


Figure 6. Cumulative distributions for the $H_r < 9.0$ subsample of objects that were seen in Figure 5. Here the Elliot et al. (2005), Gladman et al. (2012), Shankman et al. (2013), and Fraser et al. (2014) models extrapolated to $H_r = 9$ are rejectable at 99.2%, 99.1%, 82.5%, and 89.5% confidence, respectively. It is clear that these non-rejectable knee and divot models provide a better replication of the observations than either of the single exponentials, which were both rejected.

survey using the L7 model (with a single exponential absolute-magnitude distribution with $\alpha = 0.90$ extrapolated out to $H_r = 11$, where our survey's sensitivity has fallen almost to zero) until it had found 250,000 tracked detections. The AD test was used to compare our 18 plutinos to these simulated detections. Our 18 plutinos reject an L7-like model extrapolated to $H_r = 11$ at 99.1% confidence. That is, only 0.9% of the time is such an extrapolated model a worse match to 18 simulated detections than to the 18 real objects, thus rejecting the model at a highly significant level.

Because the CFEPS magnitude distribution was only constrained by plutinos in the detected range, $6.17 < H_r < 8.8$, it is not too troubling that extrapolating the magnitude distribution to $H_r = 11$ is rejectable; our study was motivated by the prior arguments that a transition near $H_r \approx 8.5$ would occur. Other works have found very different magnitude distributions for the plutinos or related populations. Elliot et al. (2005) favored a single exponential with $\alpha = 0.52 \pm 0.08$ for resonant TNOs, based on 54 resonant objects from the Deep Ecliptic Survey (DES) with $2.2 \leqslant H_r \leqslant 9.0$ (although only one object has $H_r > 8.3$ and only three have $H_r > 7.6$).[7] Shankman et al. (2013) studied 11 scattering TNOs with $6.6 \leqslant H_r \leqslant 9.5$, and found that this population favors a divot magnitude distribution, which has $\alpha_b = 0.8$ for $H_r < 8.5$, at which point the number density drops by a factor of six, followed by a shallower $\alpha_f = 0.5$ for faint objects. From an analysis of dynamically hot TNOs from various surveys (no exact number or range of magnitudes given), Fraser et al. (2014) favors a broken exponential (a knee) that has a steep exponential with $\alpha_b = 0.87^{+0.07}_{-0.2}$ for $H_r < 7.7^{+1.0}_{-0.5}$, followed by a shallower $\alpha_f = 0.2^{+0.1}_{-0.6}$ for fainter objects. Most recently, a re-analysis of DES led Adams et al. (2014) to increase their estimate of slope to $\alpha = 0.95 \pm 0.16$ for plutinos (from 31 objects with magnitudes $5.5 < H_r < 7.3$) and $\alpha = 1.04 \pm 0.19$ for Scattered objects (from 23 objects with magnitudes $5.3 < H_r < 7.2$).

As the plutinos and scattering objects are both part of the dynamically hot population, all of which are thought to share a common origin, their magnitude distributions are expected to be similar. We therefore directly tested the postulated H-magnitude distributions from Elliot et al. (2005), Shankman et al. (2013), and Fraser et al. (2014), extrapolated to $H_r = 11$, against our detections.[8] For consistency, we assume the CFEPS orbital distribution (none of the other papers derived a proposed plutino orbital parameterization) and apply the mentioned proposed absolute-magnitude distributions in the survey simulator. From ~250,000 simulated model detections in each case, we find the hypothesis that our detections are drawn from a simulated population following the Elliot et al. (2005), Shankman et al. (2013), and Fraser et al. (2014) distributions (extrapolated to $H_r = 11$) is rejectable at 93.7%, 99.5%, and 99.1% confidence, respectively; thus all models except the Elliot et al. (2005) model are highly rejectable. It is peculiar that the Elliot et al. (2005) model is not rejected, because a re-analysis of the data on which it was based (in Adams et al. 2014, by mostly authors of the original paper) indicated a far steeper slope, similar to the one in Gladman et al. (2012). The

[7] We converted the DES magnitudes from the VR-band using $g - VR = 0.4$ (B Gladman & K. Volk, 2015, private communication) and $g - r = 0.5$ as assumed for plutinos everywhere in this work, combined as $VR - r = 0.1$.

[8] We omitted testing the Adams et al. (2014) slope, because the CFEPS slope is within the uncertainty of their result.

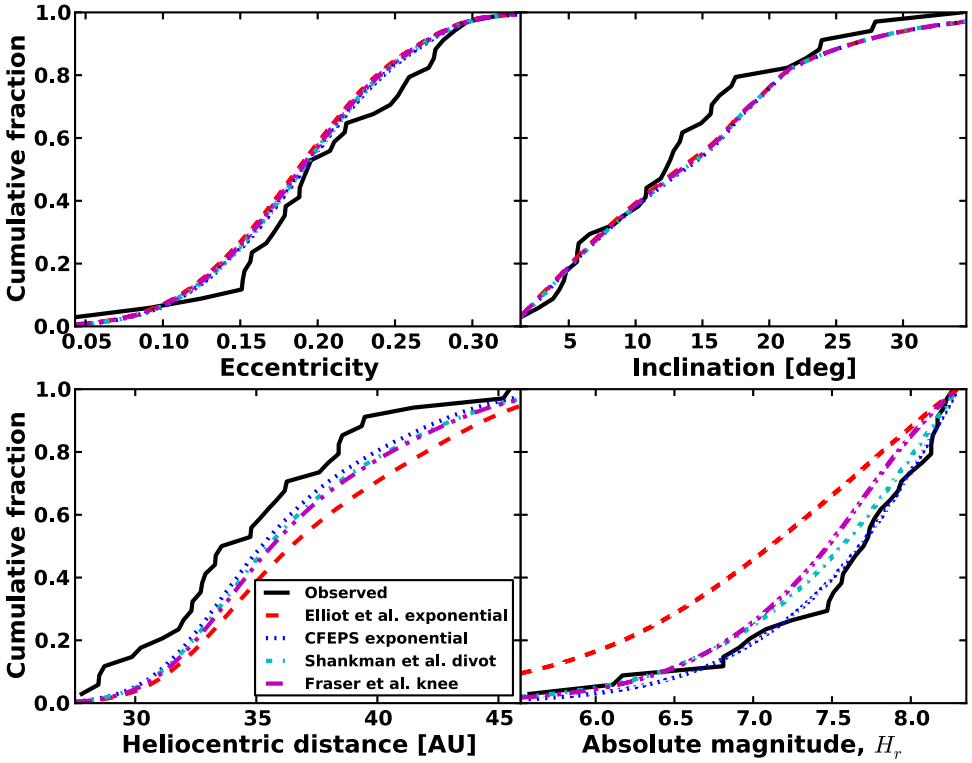


Figure 7. Cumulative distributions for the $H_r < 8.3$ subsample of the objects that were seen in Figure 5. Here the Elliot et al. (2005), Gladman et al. (2012), Shankman et al. (2013), and Fraser et al. (2014) models are rejectable at 99.98%, 56.8%, 83.6%, and 97.2% confidence, respectively. While the CFEPS exponential and the divot are not rejectable, it appears that their representation of the data would be improved by lowering the mean inclination and raising the mean eccentricity of the orbital model (raising the mean eccentricity would also improve the agreement with the heliocentric distance distribution, in the lower left).

cumulative distributions, as detected by our real and simulated survey, can be seen in Figure 4; it is clear that all of the models produce a poor representation of the data. The inclinations of the real objects are generally much lower than those of the simulated objects (independent of the model used, as the observed inclinations are only weakly dependent on the intrinsic absolute-magnitude distribution). The simulated and real detections are also in poor agreement in the observed eccentricity, heliocentric distance, and absolute-magnitude distributions.

Because we are extrapolating these model distributions far past the range of magnitudes for which they were determined, it is perhaps not surprising that these models are rejectable. When we restrict the comparison range in H-magnitude of the detections (real and simulated) to $H_r < 9.0$ (just fainter than most objects used in the cited works), our 12 remaining plutinos reject the Elliot et al. (2005), Gladman et al. (2012), Shankman et al. (2013), and Fraser et al. (2014) distributions at 94.0%, 99.94%, 94.2%, and 92.5% confidence, respectively; in other words, for $H_r < 9.0$, the divot and knee are not rejectable and present an acceptable representation of the data. It is interesting to note that our survey has no plutino detections with $8.27 < H_r < 9.01$, yet it found six plutinos (a third of the sample) with $H_r \geqslant 9.01$. Truncating the detections at $H_r = 8.3$, our remaining 12 plutinos reject the four models (in same order) at 84.8%, 78.9%, 77.2%, and 85.5% confidence. Thus, for $H_r < 8.3$, where the Shankman et al. (2013) divot is effectively just a single exponential with a slope slightly shallower than the Gladman et al. (2012) one, all of these models are non-rejectable when compared to our data. So we see that for $H_r < 8.3$, the plutino absolute-magnitude distribution is well represented by a single exponential, but when extrapolating to $H_r = 9.0$ the steep single exponential becomes rejectable, whereas when extrapolating to $H_r = 11.0$ all of the tested models become rejectable except for a shallow single exponential. In the next subsection, we combined our survey detections and characterizations with that of CFEPS and repeat the tests performed in this subsection.

4.3.2. Comparison Using 42-plutino Combined Sample

From this point onward, we combined the detections and characterizations of our survey with that of CFEPS, for a total plutino sample of 42 objects in the range $5.58 \leqslant H_r \leqslant 9.85$. As our survey was performed in r-band and CFEPS was primarily performed in g-band, some assumptions had to be made regarding $g-r$ colors when combining the surveys. Most CFEPS plutinos have measured $g-r$ colors (Petit et al. 2011) allowing simple conversion of the magnitude; for the few that did not, we converted their magnitude using the average color of the CFEPS plutinos with measured colors ($g-r = 0.5$). In the Survey Simulator, the two surveys were combined assuming a $g-r$ color distribution based on the color distribution of the CFEPS plutinos, modeled as a Gaussian distribution of width 0.2 centered on 0.5.

With this combined, 42-plutino data set, we repeated the tests performed in Section 4.3.1. For the full sample, the Elliot et al. (2005), Gladman et al. (2012), Shankman et al. (2013), and Fraser et al. (2014) models extrapolated to $H_r = 11$ are rejectable at 99.8%, 98.7%, 99.2%, and 99.4% confidence, respectively, thus rejecting all these extrapolated models (see the cumulative plots in Figure 5). With the $H_r < 9$ cut (36 real plutinos), these numbers become 99.2%, 99.1%, 82.5%, and

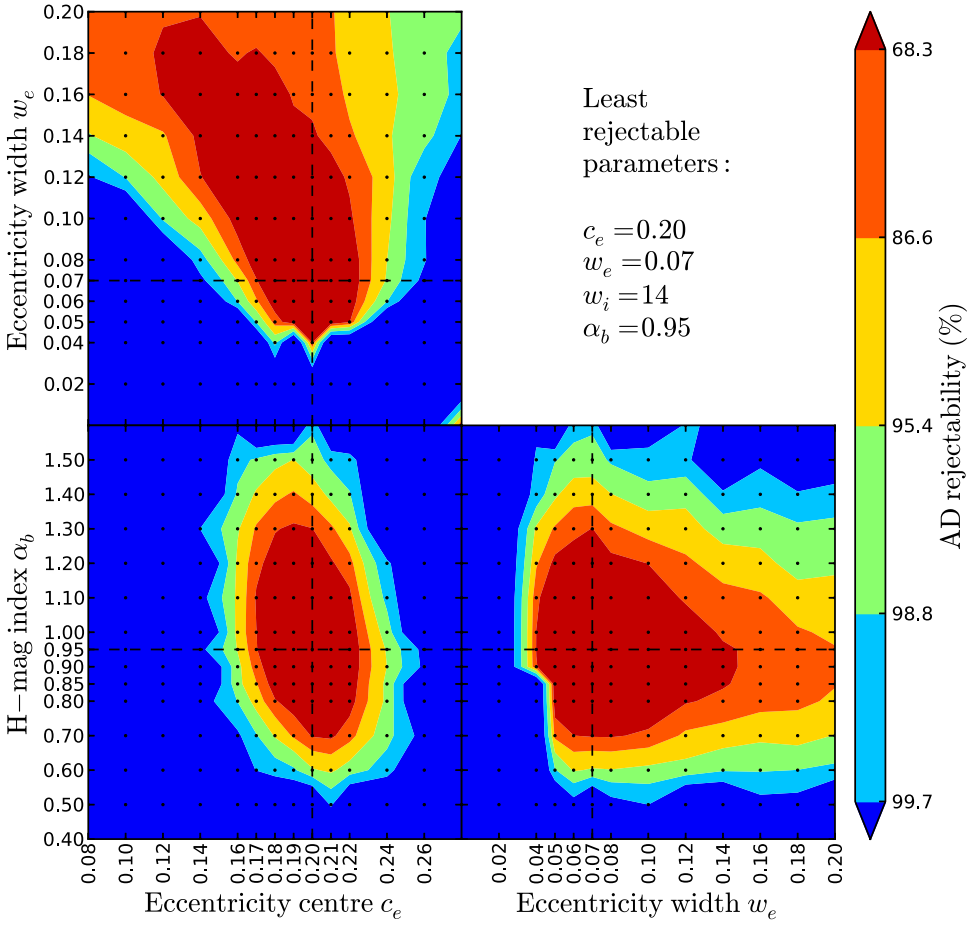


Figure 8. Rejectability in *the* c_e, w_e, and α_b space for $H_r \leqslant 8.3$. This shows the AD-rejectability of the hypothesis that our real detections with $H_r \leqslant 8.3$ could be drawn from simulated detections from models with a grid of c_e, w_e, w_i, and α_b parameters run through the survey simulator. This plot is equivalent to Figure 4 in Gladman et al. (2012) and the patterns are in good agreement, with a peak only slightly offset by values smaller than the uncertainties. Contour levels correspond to $<1\sigma$ (red), $<1.5\sigma$ (orange), $<2\sigma$ (yellow), $<2.5\sigma$ (green), $<3.0\sigma$ (cyan), and $>3\sigma$ (blue). Contours are only shown for $w_i = 14°$, because the shape of the contours are similar for all w_i, but with shrinking size when going away from $w_i = 14°$, which produces the least rejectable values. A four-dimensional grid of w_e, c_e, w_i, and α_f values was integrated. Black points indicate grid-points. The three plots are slices through the three-dimensional w_e, c_e, α_f space (with $w_i = 14°$), such that each plot is fixed to the least rejectable value of the parameter not plotted while the other two are varied.

89.5%, respectively (see Figure 6), and with the $H_r < 8.3$ cut (34 real plutinos) they are 99.98%, 56.8%, 83.6%, and 97.2%, respectively (see Figure 7). Thus, with the combination of our survey and CFEPS, we find that the Elliot et al. (2005) shallow slope is (as expected) rejectable to any of the limits tested, and the Gladman et al. (2012) steep slope is (as expected) a very good representation of the data for $H_r < 8.3$, but is highly rejectable for higher magnitudes. The reasonably steep slope of the Shankman et al. (2013) divot is also not rejectable for $H_r < 8.3$, while the Fraser et al. (2014) knee is rejectable, suggesting that the steep $\alpha = 0.8$–0.9 exponential extends nearer to $H_r = 8.3$ than the transition in the Fraser et al. (2014) knee at $H_r = 7.7$. The fact that the steep exponential is non-rejectable for $H_r < 8.3$ and highly rejectable for $H_r < 9.0$ and $H_r < 11.0$ suggests that the magnitude distribution does not extend with the same exponential form to faint magnitudes, with a drastic change in the magnitude distribution around $H_r = 8.3$. This $H_r \simeq 8.3$ transition location is very similar to the drop in number density at $H_R \simeq 8.5$ ($H_r \simeq 8.6$) that is found for the Neptunian trojans (Sheppard & Trujillo 2010b) and the $H_g \simeq 9.0$ ($H_r \simeq 8.5 \pm 0.2$) transition concluded for the scattering objects (Shankman et al. 2013). The fact that all four models are rejectable when extrapolated to $H_r = 11.0$ is unfortunate, and might mean that some of the parameters (either of the absolute-magnitude distribution or orbital distribution) need to be adjusted to provide an acceptable solution; we investigate this possibility in Section 4.4. As discussed earlier, it is doubtful that the observed transition could be due to a sudden size-dependent albedo feature.

4.4. Investigating Allowable Parameters

In Section 4.3 we showed that the single exponential H-magnitude distribution is not a valid representation for the plutinos over the entire range $5 < H_r < 11$. Meanwhile, we found that the Shankman et al. (2013) divot and Fraser et al. (2014) knee could successfully be used to at least $H_r = 9$. Thus, we now explore the allowable parameters for the knee and divot models. In addition to the allowable range of parameters for the H-magnitude distribution, we also identify the allowable range of some of the parameters of the orbital distribution. We do this partly because it is clear from Figure 4 that the 16° width of the inclination distribution is too wide, and because the detected eccentricity and H-magnitude distributions are correlated.

We split the investigation of allowable parameters into two computationally feasible sections. As the orbital distribution

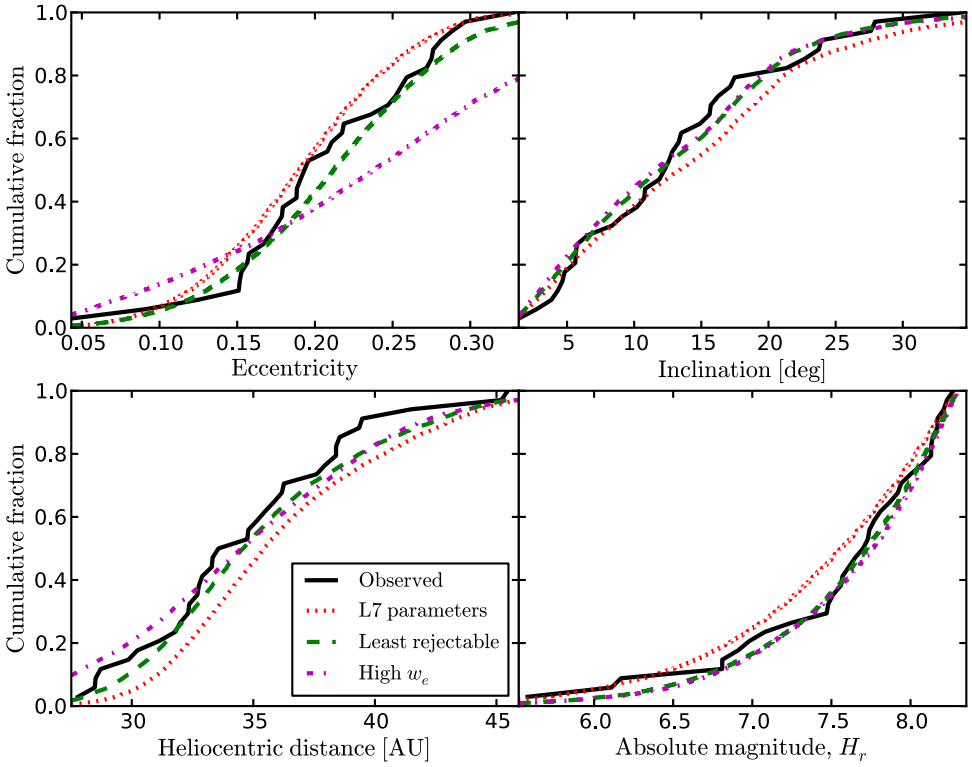


Figure 9. Cumulative distributions for different orbit distributions for objects with $H_r < 8.3$. The distribution of real detections is shown with a black solid line. Simulated detections from the least rejectable model (with $c_e = 0.20$, $w_e = 0.07$, $w_i = 14°$, *and* $\alpha_b = 0.95$) are shown with green dashes, while simulated detections from the previous L7 model (with $c_e = 0.18$, $w_e = 0.06$, $w_i = 16°$, *and* $\alpha_b = 0.90$) are shown with a red dotted line. For comparison, a very different model (on the borderline of 95% rejection confidence) is shown (magenta dash-dotted line) with $c_e = 0.20$, $w_e = 0.20$, $w_i = 14°$, and $\alpha_b = 0.95$.

should be independent of magnitude (at these sizes, there is no dynamical mechanism that would explain a size-dependent orbital distribution), we used the bright objects ($H_r < 8.3$), for which the absolute-magnitude distribution is well represented by a simple exponential, to constrain the orbital distribution and the bright slope α_b. We then fixed the orbital distribution and α_b at the least rejectable parameters, and used the full range ($H_r < 11$) of detections to constrain the location of a transition, the slope for the faint objects, and the contrast.

4.4.1. Orbital Distribution using $H_r < 8.3$ Sample

We investigated the allowable range of orbital parameters using the combined sample of plutinos with $H_r < 8.3$. This H_r cut was chosen because it only includes objects brighter than the transition, while not removing so many objects that it degrades our statistics. We also explored the allowable range of the slope parameter for the exponential H-magnitude distribution, which we have shown is valid for this range of magnitudes. For continuity, we used the familiar and accepted parameterization from CFEPS. The parameters we wished to investigate were the width w_e and center c_e for the eccentricity distribution ($e^{\frac{-(e-c_e)^2}{2w_e^2}}$ cut such that $q > 22$ au), the width w_i of the inclination distribution ($\sin(i)e^{\frac{-i^2}{2w_i^2}}$), and the exponent α_b of the exponential H-magnitude distribution ($10^{\alpha_b H_r}$), where the b subscript denotes that we were only investigating the bright objects (brighter than any transition). These model distributions do not come from cosmogonic motivation, but CFEPS empirically found them to satisfactorily represent the detections once weighted by the detection biases (Gladman et al. 2012). Given that these models, with the right parameters, continue to be non-rejectable (see Figure 8), their functional forms are still appropriate.

We ran survey simulations using a four-dimensional grid of the c_e, w_e, w_i, α_b model parameters. CFEPS and our new survey combined have 34 plutinos with $H_r < 8.3$. From this analysis, we found that the L7 model parameters are near the center of the allowable range. Figure 8 shows contour plots showing the rejectability of various parameters. It can be seen that for a given value of the eccentricity width, α_b and c_e have a clearly defined allowable region. The eccentricity width w_e, however, is not well constrained, with non-rejectable values extending beyond 0.20. It might seem odd that such a high-eccentricity width is allowable, together with the high-eccentricity center ($c_e \sim 0.2$), but as our model has a cut where $q > 22$ au, the high-eccentricity tail that would make such a combination of w_e and c_e rejectable does not exist. Values of α_b up to 0.5 (as in Elliot et al. 2005) can be rejectable for any combination of the other parameters, while values up to 1.50 are borderline acceptable in some cases. The eccentricity center c_e is the best constrained (smallest fractional allowable range), as long as w_e is low; if w_e is increased, the allowable range of c_e increases, primarily by smaller values of c_e becoming non-rejectable. An important correlation remains that a lower-eccentricity center requires a larger width w_e to produce the copious numbers of detected $e > 0.2$ plutinos.

The allowable parameters found from this analysis are $c_e = 0.20^{+0.04}_{-0.04}$, $w_e = 0.07^{+0.15}_{-0.03}$, $w_i = 14°^{+7°}_{-3°}$, and $\alpha_b = 0.95^{+0.45}_{-0.35}$ (center values represent the least rejectable value, uncertainties represent the one-dimensional ranges for which the rejectability

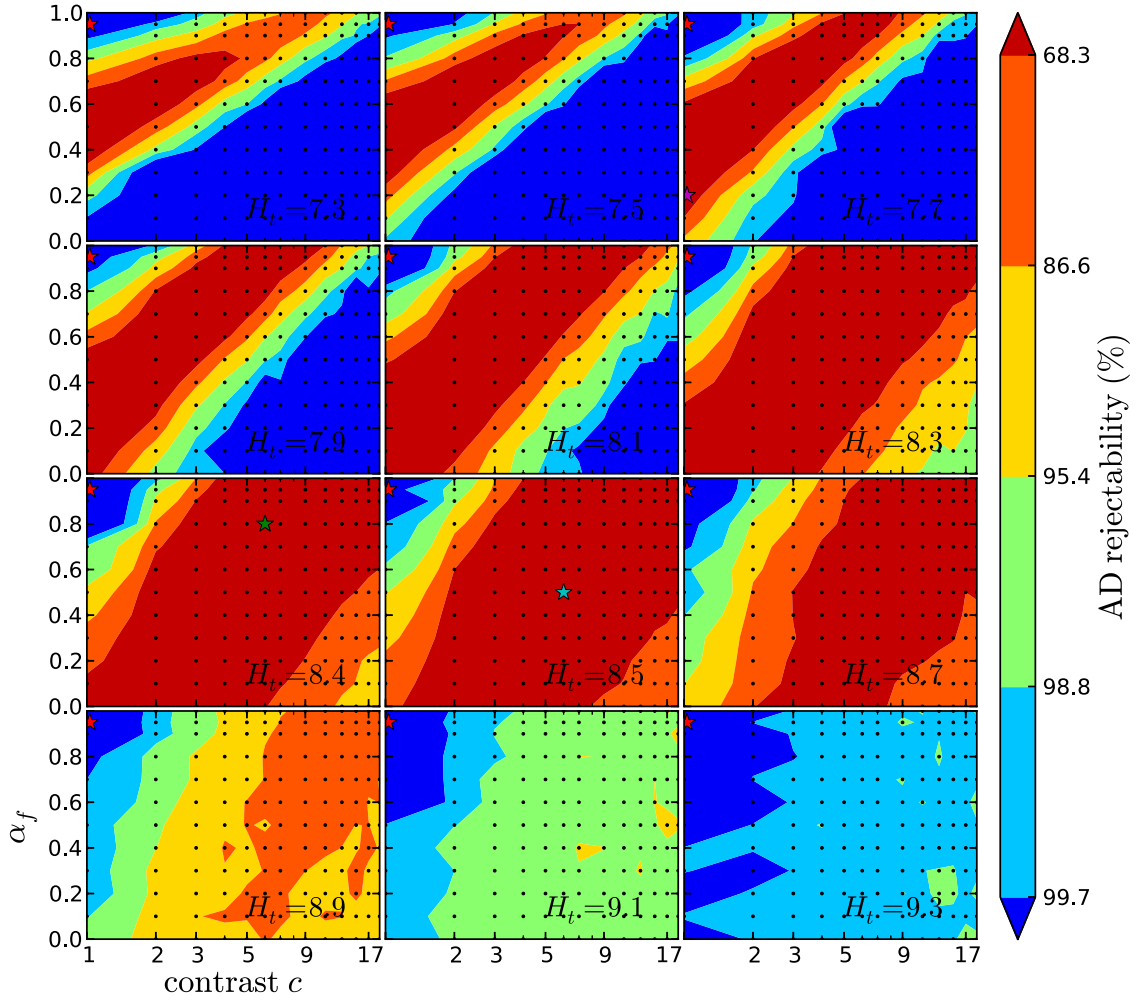


Figure 10. Rejectability in the H_t, c, α_f space, with $\alpha_b = 0.95$ and the least rejectable orbital parameters from Section 4.4.1. Each subplot has the same axis and is a slice through the three-dimensional space at a particular value of H_t (labeled on each subplot). Colors are the same as in Figure 8; models that are *not* rejectable at 95% confidence are shown in red, orange, and yellow. With $H_t > 9$, all solutions are rejectable because the steep α_b cannot continue to $H_r = 9$; there must be a transition at a brighter magnitude. For $H_t < 8.8$, knees (solutions with $c = 1$) are not rejectable at 95% confidence. Note that knees to $\alpha_f \geqslant 0.8$ are always rejectable at 95% confidence; for $H_t > 8.2$, knees to $\alpha_f \geqslant 0.6$ are also rejectable. The least rejectable set of parameters is marked with a green star ($H_t = 8.4$ panel). Also marked are the knee parameters favored by Fraser et al. (2014) for hot TNOs (magenta star; $H_t = 7.7$ panel), the divot parameters favored by Shankman et al. (2013) for scattering TNOs (cyan star; $H_t = 8.5$ panel), and the single exponential model (red star on every plot, as varying H_t when $c = 1$ and $\alpha_f = \alpha_b$ changes nothing).

is <95% *when all other parameters are fixed*). The previous CFEPS best estimates were $c_e = 0.18^{+0.03}_{-0.04}$, $w_e = 0.06^{+0.03}_{-0.02}$, $w_i = 16^{\circ+8^{\circ}}_{-4^{\circ}}$, and $\alpha_b = 0.90 \pm 0.25$. It is noteworthy that despite this analysis using a factor ∼2 more objects, our constraint of the allowable range of parameters is not improved. This is due to the difference in the statistic used here, where our method is more rigorous but less constraining. The steep slope of $\alpha_b = 0.95^{+0.45}_{-0.35}$ (95% confidence), constrained using the range $5.5 < H_r < 8.3$, is nearly identical to recent results from the re-analyses of DES (Adams et al. 2014), which measured $\alpha_b = 0.95 \pm 0.16$ (1σ) over the range $5.5 < H_r < 7.3$. Figure 9 presents a visual representation of the simulated and real detections illustrating the differences between the CFEPS parameters and the least rejectable parameters found here.

4.5. Magnitude Distribution for $H_r < 11$

Making the reasonable base assumption that the orbital distribution is not dependent on absolute magnitude, we fixed the plutino orbital distribution to the least rejectable parameters from Section 4.4.1. This allowed us to explore the remaining H-magnitude space to determine what it could reveal about the absolute-magnitude distribution. We implemented the four parameter divot H-magnitude distribution as parameterized by Shankman et al. (2013), which turns into a knee when $c = 1$ and further becomes a single exponential when $c = 1$ and $\alpha_b = \alpha_f$. We fixed $\alpha_b = 0.95$ (from the least rejectable parameter-set in Section 4.4.1), and investigated the remaining three-dimensional parameter space (H_t, c, α_f) for plutinos with $H_r < 11$, thus comparing to the full set of 42 detected plutinos.

Figure 10 shows contour plots of the rejectability of the explored H_t, c, and α_f combinations. For all values of H_t a correlation exists for which increasing the contrast requires an increase in α_f. This is because our survey found six plutinos with $H_r > 9$, so the distribution must have sufficient $H_r > 9$ objects to explain these detections; if there is a deep contrast, the post-divot slope must be steep to generate the $H_r > 9$ objects. At a given H_t, knees thus have the smallest allowable α_f, and the maximum allowable α_f is also lowest for knees. For the brightest possible transition magnitude we investigated (H_t

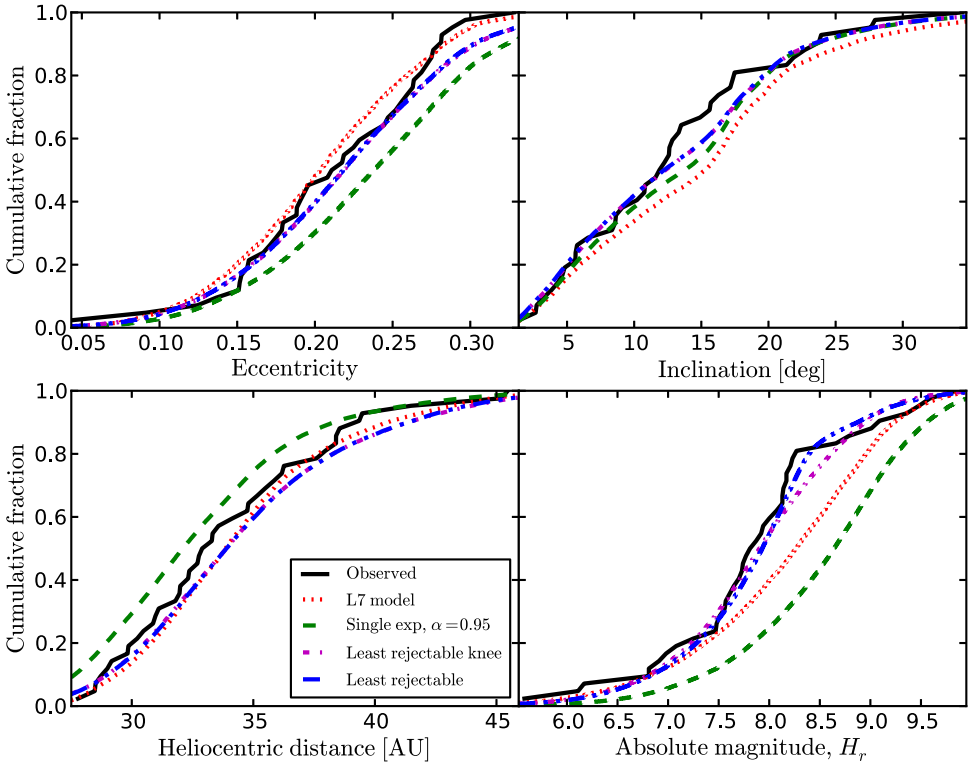


Figure 11. Cumulative distributions of the least rejectable models, for the full combined plutino sample. Our simulations generated objects down to $H_r = 11$, which is well beyond our faintest real detection of 9.85, to prevent artificially cutting the simulated detections. The least rejectable divot (blue dash double-dotted line) had $H_t = 8.4$, $c = 6$, and $\alpha_f = 0.80$, while the least rejectable knee (magenta dash-dotted line) had $H_t = 7.7$, $c = 1$, and $\alpha_f = 0.40$. It is clear that both of these solutions provide a better representation of the observations than the single exponential of same α_b (green dashed line) and the L7 model (with original orbital and magnitude parameters; red dotted line) over this full magnitude range. This is not surprising, given that our survey detected no plutinos with $8.27 < H_r < 9.01$, yet found six plutinos with $H_r \geqslant 9.01$.

$= 7.3$), a range of divot and knee parameters are non-rejectable; a knee requires a faint slope in the range $0.2 \leqslant \alpha_f \leqslant 0.8$, while as the transition contrast c increases, the range of allowable α_f values gets narrower and shifts to higher values. As H_t increases, the smallest non-rejectable α_f decreases (or inversely, a deeper contrast c becomes allowable for a given α_f). Deep contrasts are allowable as long as the post-divot slope is steep; this is because of the dearth of detections with $8.3 < H_r < 9.0$ in our sample, combined with the six detections with $H_r > 9.0$. Eventually, the maximum allowed α_f falls with increasing H_t, especially for the knee; the faintest location for a knee to be rejectable at less than 95% confidence is between $H_t = 8.7$–8.9; for $H_t > 8.9$, a knee is always rejectable. With the full set of data, we find that the least rejectable set of parameters is for a moderately deep divot ($c = 6$) at $H_t = 8.4$ to a moderately steep slope ($\alpha_f = 0.8$ for $H_r = 8.4$–11.0) near the Shankman et al. (2013) parameters, while the least rejectable knee with $H_t = 7.7$ and $\alpha_f = 0.4$, is very near the Fraser et al. (2014) parameters. Figure 11 shows the cumulative distribution of real and simulated detections from various models.

A steep ($\alpha_f > 0.6$) post-transition slope cannot continue because if such a slope is extrapolated indefinitely, the total mass of the trans-Neptunian region would diverge (Chiang & Brown 1999; Gladman et al. 2001). So, an $\alpha_f \simeq 0.8$ post-divot slope can only be suitable over a limited H-magnitude range; there would have to be another transition at smaller sizes to a gentler slope. The main asteroid belt is seen to have multiple gentle transitions in the known size distribution; it is quite possible that TNOs would have a similar pattern. Alternately, Figure 10 makes it clear that many $\alpha_f < 0.6$ divot and knee models are not rejectable, including the original divot ($H_t = 8.5$, $c = 6$, $\alpha = 0.5$) and knee ($H_t = 7.7$, $c = 1$, $\alpha = 0.2$) proposed by Shankman et al. (2013) and Fraser et al. (2014), respectively; it is thus entirely possible that the plutinos feature an $\alpha_f < 0.6$ slope starting from a $H_r \sim 8$ transition.

4.6. Population Estimate

CFEPS predicted 13000^{+6000}_{-5000} plutinos with $H_g < 9.16$ (Gladman et al. 2012), corresponding to $H_r < 8.66$, where uncertainties indicate the 95% confidence range. Our survey combined with CFEPS has 34 plutinos with $H_r < 8.66$, so we ran 10,000 survey simulations of the combined surveys until each simulation detected 34 synthetic objects with $H_r < 8.66$. We recorded the total number of generated objects required to provide these detections for each run to estimate the total plutino population size and uncertainty (all population estimates in this work assume a uniform prior). This was done for two absolute-magnitude distributions: the least rejectable knee and the least rejectable divot found above. With this bootstrap approach, we estimate 9000 ± 3000 plutinos with $H_r < 8.66$, regardless of whether we use the divot magnitude distribution or the knee distribution, showing a $\sim\frac{1}{\sqrt{2}}$ improvement in relative uncertainty as expected from roughly doubling the number of detections. This revised population estimate is in agreement with the CFEPS prediction, although this is not too surprising due to the partial overlap of the sample.

Figure 1 of Adams et al. (2014) shows that they estimate ~3000 plutinos with $H_{VR} \leqslant 8.1$ ($H_r = 8.0$, as $VR - r = 0.1$,

see Footnote 7). With the divot magnitude distribution, which for $H_r < 8.4$ is simply a single exponential with $\alpha = 0.95$, we estimate 8000^{+3000}_{-2000} plutinos with $H_r < 8.4$. Scaling this down, our estimate corresponds to 3300^{+1300}_{-800} plutinos with $H_r < 8.0$. The observations of Adams et al. (2014) are thus in good agreement with our own population estimates and those of CFEPS.

The addition of the new, deeper survey blocks described in this work allows us to estimate the population of even fainter objects (Table 4). We estimate the existence of 37000^{+12000}_{-10000} plutinos with $H_r < 10.0$ assuming the divot distribution, and 35000^{+12000}_{-10000} if the knee is assumed. It may seem surprising that these two models produce such similar population estimates because they have such different values of α_f. However, this is because the proposed knee occurs at a brighter magnitude than the divot, so the divot has more objects with $7.7 < H_r < 8.4$ than the knee, whereas the knee has more objects just past $H_r = 8.4$, thus roughly equalizing the total population estimate. If one could probe several magnitudes past the transition, the difference in population would become clearer. These population estimates show that despite having transitioned away from a single exponential absolute-magnitude distribution, the plutinos continue to significantly outnumber the inner Solar System's asteroid belt, which has 592 objects with $H_r < 10$.[9]

5. NEPTUNIAN AND URANIAN CO-ORBITALS

Co-orbital objects are in the 1:1 mean-motion resonance with a planet (thus having, on average over time, the same orbital period and semimajor axis as the planet) and a librating resonant angle $\phi_{11} = \lambda - \lambda_P$, where ϕ_{11} approximately measures how far ahead of the planet an object is in orbital phase. For co-orbital motion, ϕ_{11} librates around one of four values: 0°, 60°, 180°, and 300° (Mikkola et al. 2006). *Quasi-satellites* librate around 0°; in the co-orbital frame these move like retrograde satellites, despite being outside the planet's gravitational dominance. Leading and trailing *trojans* librate around the L4 (60° ahead of planet) and L5 (300° ahead =60° behind planet) Lagrange points, respectively. Objects on *horseshoe orbits* execute librations around L3 (180° from the planet) with high-amplitudes that encompasses the L3, L4, and L5 Lagrange points.

Co-orbital objects can be further divided into two classes: stable co-orbitals on long-term stable orbits that keep them co-orbital for the age of the Solar System and temporary co-orbitals that leave the co-orbital motion on timescales significantly shorter than the age of the Solar System. Of the different libration modes, only trojans are generally long-term stable (resonant on Gyr timescale; Mikkola et al. 2006; Ćuk et al. 2012). Temporary co-orbitals are almost certainly scattering objects temporarily captured into co-orbital motion (Morais & Morbidelli 2002, 2006; Alexandersen et al. 2013a). As such, they likely have completely different histories and origins from the stable co-orbitals, aside from having a different orbital distribution. We therefore treated the stable and temporary co-orbitals as separate populations.

There are currently 14 known Neptunian trojans,[10] of which two are certain to be temporary (Horner & Lykawka 2012, this work) and two have uncertain stability (Brasser et al. 2004; Horner & Lykawka 2010; Guan et al. 2012; Horner et al. 2012). These 14 were discovered in several different surveys, some of which do not have publicly available characterization, making it impossible to analyze the combined set reliably. The following analysis therefore only used detections from our survey and CFEPS. Our survey found one temporary Uranian co-orbital (mal01 = 2011 QF_{99}) and two Neptunian co-orbitals (one stable trojan and one temporary trojan, mah02 = 2012 UV_{177} and mah01 = 2012 UW_{177}, respectively). CFEPS (Petit et al. 2011) found one temporary Neptunian co-orbital (2004 KV_{18}), which was recognized as a temporary trojan by Horner & Lykawka (2012).

Normally, population estimates based on so few detections are highly uncertain and depend strongly on the choice of toy model orbit distributions used to make the population estimate. However, in this case we have access to independent estimates of the orbital distributions of the temporary Uranian and Neptunian co-orbitals (theoretical work in Alexandersen et al. 2013a), and the stable Neptunian trojans (i.e., the statistical analysis of all known Neptunian trojans in Parker 2015). Our analysis based on these models is detailed below. Assuming that the dynamically hot TNO populations share a common origin and thus have the same size distribution (Shankman et al. 2013), the distinction between stable and temporary Neptunian trojans is simply the difference in the orbital distributions; however, we will continue to treat them separately for this work. Population estimates of the three classes of co-orbitals treated in this work are presented in Table 4, assuming the magnitude distributions of other hot populations found in the literature.

5.1. Stable Neptunian Trojans

Our survey discovered one long-term stable trojan of Neptune (mah02 = 2012 UV_{177}), with $H_r = 8.93$ in the high-lat block of our new survey. Via 4 Gyr numerical integrations, using the range of orbits established by the orbital classification algorithm of Gladman et al. (2008, p. 43), we find this to be a secure stable 1:1 trojan resonator, with a libration amplitude around the L4 point of 13° (see Figure 12). The fact that 2012 UV_{177} has an inclination of 21° and that we discovered no stable Neptunian trojans in our low-lat block, which is biased against detecting high-inclination objects, continues to strengthen the assertion that the stable Neptunian trojans have a very dynamically hot inclination distribution (Sheppard & Trujillo 2006, 2010a). Recent studies indicate that the standard 15° width (of other hot populations) is still plausible, although at the low end of the allowable range (Parker 2015).

CFEPS did not discover any long-term stable trojans. We use our one detection and the characterization of both surveys to derive order-of-magnitude population estimates using orbital and absolute-magnitude distributions derived elsewhere in literature. While the detection of only one object means that the lower limit of the population estimate is uncertain and essentially consistent with 0 (or 1), the value of these estimates

[9] The main asteroid belt (2.06 au < a < 3.65 au) is completely known for $H < 15.75$ (Ryan et al. 2015). We have thus counted the number of objects with 2.06 au < a < 3.65 au and $H_r < 10$ in the MPC database file MPCORB.DAT from http://www.minorplanetcenter.net/iau/MPCORB.html. By comparing the magnitudes of the objects from our survey, we found that MPCORB.DAT had a $H - H_r \approx 0.20$ offset, which we accounted for.

[10] This work in addition to the list of Neptunian trojans in the MPC database as of 2015 May 9: ⟨http://web.archive.org/web/20150509015658/http://www.minorplanetcenter.org/iau/lists/NeptuneTrojans.html⟩.

Table 4
Population Estimates for All Populations Investigated in This Work

	H-model	$N(H_r \leqslant 8.4)$	$N(H_r \leqslant 8.66)$	$N(H_r \leqslant 9.1)$	$N(H_r \leqslant 10.0)$	$N(H_r \leqslant 11.0)$
Temporary	Knee	$\leqslant 300$	$\leqslant 300$	$\leqslant 300$	110^{+400}_{-100}	190^{+800}_{-180}
Uranian co-orbitals	Divot	$\leqslant 300$	$\leqslant 300$	$\leqslant 300$	110^{+500}_{-100}	270^{+1200}_{-260}
Temporary	Knee	70^{+300}_{-60}	70^{+300}_{-60}	90^{+400}_{-80}	200^{+900}_{-190}	1200^{+3000}_{-1000}
Neptunian co-orbitals	Divot	70^{+300}_{-60}	80^{+300}_{-70}	90^{+400}_{-80}	210^{+900}_{-200}	2500^{+6000}_{-2100}
Stable	Knee	$\leqslant 250$	$\leqslant 250$	80^{+300}_{-70}	140^{+600}_{-130}	...
Neptunian Trojans	Divot	$\leqslant 250$	$\leqslant 260$	80^{+300}_{-70}	150^{+600}_{-140}	...
Plutinos	Knee	7000^{+3000}_{-2000}	9000 ± 3000	14000^{+5000}_{-4000}	35000^{+12000}_{-10000}	...
	Divot	8000^{+3000}_{-2000}	9000 ± 3000	12000^{+4000}_{-3000}	37000^{+12000}_{-10000}	...
	Exponential	$\geqslant 530$	$\geqslant 890$	$\geqslant 2300$	...	...
3:1 ($0.40 \leqslant e \leqslant 0.57$)	Knee	$\geqslant 750$	$\geqslant 1200$	$\geqslant 2000$	...	...
	Divot	$\geqslant 950$	$\geqslant 1100$	$\geqslant 1300$	...	...
	Exponential	$\geqslant 16$	$\geqslant 26$	...	...	...
4:1 ($0.49 \leqslant e \leqslant 0.65$)	Knee	$\geqslant 18$	$\geqslant 24$	...	...	...
	Divot	$\geqslant 17$	$\geqslant 27$	...	...	...

Note. Medians are given with 95% confidence ranges (2.5% in each tail); 95% confidence upper limits (5% chance of higher values) when based on zero real detections. For the 3:1 and 4:1 resonances, 95% confidence lower limits (5% chance of lower values) are given due to the lack of model constraints and the small number of real detections. For trojans and co-orbitals, orbital distributions from the literature were used (Alexandersen et al. 2013a; Parker 2015); estimates are given for two different absolute-magnitude distribution models found in the literature for related populations, the Fraser et al. (2014) knee with $\alpha_b = 0.87$, $H_t = 7.7$, and $\alpha_f = 0.20$, and the Shankman et al. (2013) divot with $\alpha_b = 0.8$, $H_t = 8.5$, $c = 6$, and $\alpha_f = 0.50$. For plutinos, the least rejectable orbital parameters from Section 4.4.1 were used; population estimates are presented for the least rejectable knee ($\alpha_b = 0.95$, $H_t = 7.7$, and $\alpha_f = 0.40$) and divot ($\alpha_b = 0.95$, $H_t = 8.4$, $c = 6$, and $\alpha_f = 0.80$) from Section 4.5. For the 3:1 and 4:1 resonances, toy models with a restricted and uniform eccentricity distributions (Section 6) were used to generate lower limits; lower limits are presented using the Fraser et al. (2014) knee, the Shankman et al. (2013) divot, and an $\alpha = 0.9$ single exponential.

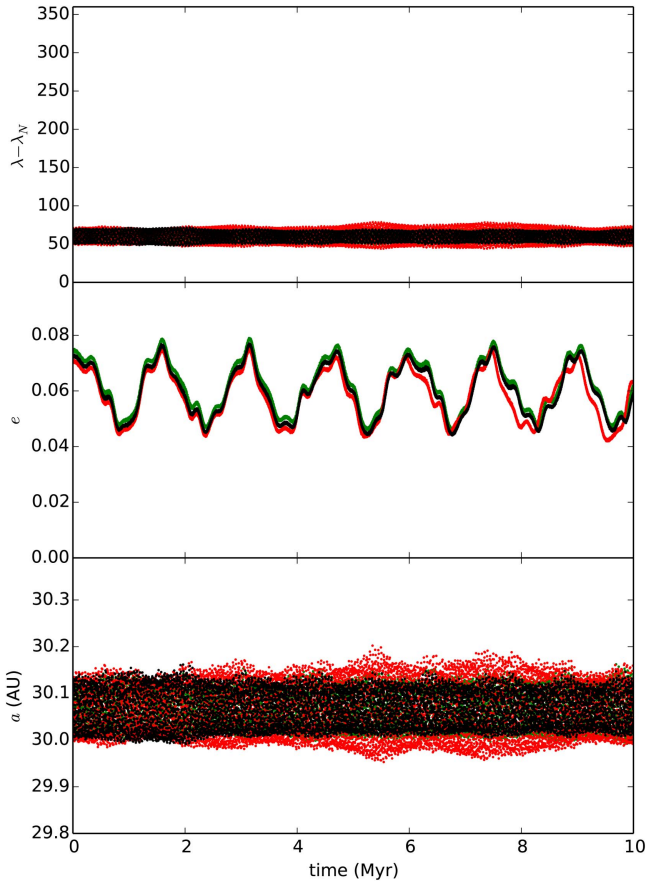


Figure 12. Future evolution of 2012 UV_{177}, the stable Neptunian trojan. The numerical integration of the best-fit orbit (black) is shown, as well as the smallest and largest semimajor axis orbits compatible with the astrometric residuals (green and red, respectively). All three clones remain as L4 trojans for the 10 Myr integration used in classification. A 4.5 Gyr integration shows that these orbits remain stable for the age of the Solar System.

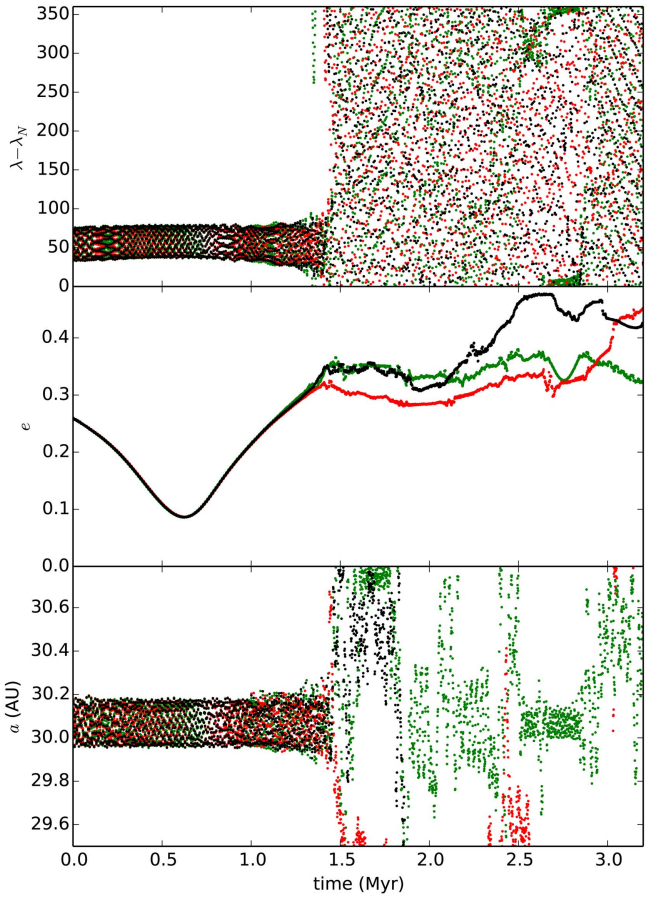


Figure 13. Future evolution of 2012 UW_{177}, the temporary Neptunian trojan. The numerical integration of the best-fit orbit (black) is shown, as well as the smallest and largest semimajor axis orbits compatible with the astrometry (green and red, respectively). All three clones remain as L4 trojans for about 1.4 Myr, at which point they escape from the resonance. The min-a orbit briefly gets trapped as a quasi-satellite between 2.5 and 2.8 Myr from now.

lies in the fact that the non-detection of more objects sets a strong upper limit.

We used the orbital distribution of the stable Neptunian trojans derived by Parker (2015). As we cannot constrain the absolute-magnitude distribution of the trojans, we use distributions as found for other related populations in the literature (the Shankman et al. 2013 divot and Fraser et al. 2014 knee); we did not use a steep single exponential, as that has already been rejected for the trojans (Sheppard & Trujillo 2010b). Sheppard & Trujillo (2010b) estimated about 400 Neptunian trojans (stable + temporary) with $D > 100$ km, which corresponds to $H_r < 8.6$, given their assumptions and appropriate color conversion. CFEPS (Gladman et al. 2012) put an upper limit of 300 (95% confidence) stable Neptunian trojans with $H_g < 9.16$ (chosen to correspond with $D > 100$ km), which corresponds to $H_r < 8.66$ (assuming colors similar to the plutinos). Our 95% confidence upper limit of <250 trojans (L4 +L5) with $H_r < 8.66$ is even lower than previous estimates and disagrees with the Sheppard & Trujillo (2010b) estimation of ~400. However, part of this difference is because only seven of the nine objects on which Sheppard & Trujillo (2010b) based their population estimate turned out to be trojans; additionally, one of the objects is a temporary trojan.

Our estimated cumulative populations of stable trojans for $H_r < 9.1$ and <10.0 are 80^{+300}_{-70} and 150^{+600}_{-140}, respectively (these numbers are for a divot H distribution, but the knee does not produce significantly different population estimates; see Table 4). Even at these higher magnitude cuts, our population estimates are lower than the Sheppard & Trujillo (2010b) prediction. Stable Neptunian trojans are thus not nearly as numerous as the initial estimates, and almost certainly less numerous than the main asteroid belt, which has 592 asteroids with $H_r < 10.0$ (See Footnote 9).

5.2. Temporary Neptunian Co-orbitals

Our survey found one temporary Neptunian co-orbital: 2012 UW_{177}. With $e = 0.259$ and $i = 53\overset{\circ}{.}89$, 2012 UW_{177} has the most extreme orbit of the Neptunian co-orbitals discovered to date. At $H_r = 10.65$ it is also the intrinsically faintest (and so likely smallest); assuming 5% albedo, its diameter is ≈45 km (or ≈25 km for a 16% albedo). Integrations show that although this object is currently librating around Neptune's leading L4 Lagrange point (executing motion 35°–75° ahead of Neptune), in less than 1.5 Myr the long-term cycle of its eccentricity (caused by secular interactions with the other giant planets) will cause it to dynamically couple with Uranus. Because the 1:1 resonance with Neptune provides no protection against Uranus encounters, the object is then perturbed out of the resonance to scattering behavior (see Figure 13). Thus it appears plausible that as recently as 200 kyr ago (when our integrations show the eccentricity was sufficiently high for Uranus coupling) the object was a scattering object (or Centaur if $a < 30$ au) that had a Uranus encounter, which perturbed the orbital semimajor axis to near that of Neptune, allowing the 1:1 resonance dynamics to begin. The resonance dynamics protected the object from Neptune encounters, while a secular cycle in the pericentric distance q protected the object from Uranus for a Myr (q is currently rising through 22.3 au to a maximum near 28 au about 400 kyr from now). However, our integrations have all clones re-coupling to Uranus during the next eccentricity maximum about 1.5 Myr from now and exit the resonance. The past and future behavior of this temporarily captured object is similar to that of 2011 QF_{99}, with the only noteworthy feature being the unusually large orbital inclination. Due to the large planetary encounter speeds produced by the large i, the perturbations by both Uranus and Neptune are weak, resulting in small (a, q, i) changes in any single encounter; this implies that this small body was likely originally delivered to $a \simeq 30$ au with $i > 50°$. The source of these large-inclination TNOs remains unclear (Gladman et al. 2009a; Brasser et al. 2012; Volk & Malhotra 2013).

CFEPS also discovered one temporary Neptunian co-orbital, 2004 KV_{18}, with $H_r \sim 8.3$ (Petit et al. 2011). 2004 KV_{18} was the first Neptunian trojan shown to definitely be temporary; it is currently on an L5 trojan orbit that is unstable on ~0.2 Myr timescale (Horner & Lykawka 2012).

Alexandersen et al. (2013a) simulated the capture of temporary Neptunian and Uranian co-orbitals from the scattering/centaur population, finding a theoretical orbital distribution of these objects. As the temporary co-orbitals from our surveys are insufficient to empirically constrain the orbital distribution, we used this theoretical distribution to calculate population estimates (Table 4). We applied absolute-magnitude distributions in the form of the knee (Fraser et al. 2014) and the divot (Shankman et al. 2013) distributions for other closely related populations. As with the stable trojans, we omitted the single exponential, because this is rejectable for temporary co-orbitals and the scattering/centaur populations (Alexandersen et al. 2013a; Shankman et al. 2013). Surprisingly, the number of stable Neptunian trojans and temporary co-orbitals appears to be very similar. The dynamical lifetimes of the temporary co-orbitals are less than 0.1% of the age of the Solar System. Given similar population numbers, it is therefore impossible for all the temporary Neptunian co-orbitals to be previously stable Neptunian trojans diffusing from the stability zone. Given the lifetime ratio of at least 1000:1 for stable:temporary co-orbitals, at most ~0.1% of co-orbitals should be seen as temporary in a steady-state scenario. Even if the true populations of the stable and temporary co-orbitals were in the 95% upper and lower tail of our predictions (380 stable and 10 temporary co-orbitals with $H_r < 9.1$), the implication that $\gtrsim$2.5% of Neptunian co-orbitals are unstable on timescales of 0.1% of the age of the Solar System is significantly above what is allowable in steady state. The temporary co-orbitals therefore must be recently (and continuously) captured from a different reservoir, such as the scattering/centaur population (Alexandersen et al. 2013a). The large inclination of 2012 UW_{177} should not be taken as an indication of the inclination distribution of the stable Neptunian trojans. In the scenario where they are captured from the scattering/Centaur population, the temporary Neptunian co-orbitals could have a much hotter inclination distribution than the stable Neptunian trojans.

5.3. Temporary Uranian Co-orbitals

Our survey discovered one temporary Uranian co-orbital, mal01 = 2011 QF_{99} (Alexandersen et al. 2013a). CFEPS did not discover any Uranian co-orbitals. With only one temporary Uranian co-orbital in our sample, we made an order of magnitude population estimate of temporary Uranian co-orbitals, again based on the orbital distribution computed in Alexandersen et al. (2013a) and the H distributions from Shankman et al. (2013) and Fraser et al. (2014), as in Section 5.2. See Table 4 for population estimates at various H cuts. While the large uncertainty ranges overlap, the most likely

estimates suggest that there are fewer temporary Uranian co-orbitals than Neptunian ones, which is in agreement with the theoretical prediction that in steady state, the temporary Neptunian co-orbitals should outnumber the temporary Uranian co-orbitals by a factor of about seven (Alexandersen et al. 2013a).

6. 3:1 AND 4:1 RESONATORS—POPULATION ESTIMATES

Our survey discovered three objects (mal23 = 2011 UR_{411}, mal08 = 2011 US_{411}, and mal62 = 2011 UQ_{411}) in the 3:1 outer mean-motion resonance with Neptune ($a \approx 62.5$ au), and one object (mal33 = 2011 UP_{411}) in the 4:1 resonance ($a \approx 75.8$ au). These resonances have been very sparsely studied in the observational literature; in fact, 2011 UP_{411} appears to be the first known object to be securely determined to be in the 4:1 resonance. Theoretical studies (Beauge 1994; Malhotra 1996) have provided some insight into the dynamical structure of the resonances, and we created some simple toy models of the 3:1 and 4:1 orbital distributions. Due to the large number of unknown parameters, the small sample, and our lack of sensitivity to low-eccentricity objects (as only very large objects, of which there are few, would be sufficiently bright to be detectable on near-circular orbits at that distance), we have taken care to present very conservative lower-limit population estimates. For these population estimates, we used a single exponential H-magnitude distribution with $\alpha = 0.9$, as slopes of 0.8–1.0 are typically seen for $H_r < 8$ for dynamically hot TNO populations and have not been rejected by these populations. However, as the single exponential has been rejected by multiple other dynamically hot populations, we also produce population estimates using alternative absolute-magnitude distributions proposed in the literature for other hot populations (the Shankman et al. 2013 divot and the Fraser et al. 2014 knee).

In addition to the three objects in the 3:1 resonance discovered by our survey, CFEPS discovered one: 2004 VU_{130} = L4V08.[11] This combined sample of four 3:1 objects spans $0.40 \leqslant e \leqslant 0.50$, $8^\circ \leqslant i \leqslant 40^\circ$ and $6.2 \leqslant H_r \leqslant 9.1$. We used the simple CFEPS L7 model (Gladman et al. 2012) of the 3:1 resonance to determine a population estimate. Four objects is insufficient to constrain the orbital distribution significantly, and the population estimate will be quite sensitive to the choice of model. For the temporary and stable co-orbitals (Section 5), we had access to independently determined orbital distributions, but for the 3:1 and 4:1 distribution we do not have this benefit, because we present some of the first discoveries. Due to the strong model-dependence, we sampled a range of plausible models.

As our surveys were mostly sensitive to very eccentric orbits, we have no knowledge of whether low-eccentricity 3:1 objects exist. For that reason, we constrained the eccentricity distribution such that we only produced a population estimate for objects on orbits that are at least as detectable as the observed ones; the eccentricity of synthetic objects was therefore constrained to a flat distribution with $0.40 \leqslant e \leqslant 0.57$. The upper limit of $e \leqslant 0.57$ was chosen to produce only objects with $q > 27$ au, as very few known stable TNOs have smaller pericenter distances. By using a flat eccentricity distribution and extending it to 0.57, we ensure that our population estimate will be on the low side (as resonances typically have fewer objects as the upper e limit is approached). We used a flat semimajor axis distribution from 62.1 to 62.5 au. Population estimates were produced for a range of inclination widths from $w_i = 5^\circ$ to 30° in 5° increments, and for a symmetric fraction of 10% to 90% in 20% increments (without greater statistics or a theory of the mechanism that populated the resonance, the symmetric fraction is unknown).[12] A simple one-dimensional AD test ruled out $w_i = 5^\circ$ and $w_i = 10^\circ$ at more than 95% confidence in every case; we only used models with a non-rejectable inclination distribution to produce our population estimate. From these integrations, we computed the 95% confidence lower limit for each of the non-rejectable models; the lowest of these was chosen as our final lower-limit population estimate.[13] We found that the three different absolute-magnitude models (exponential, knee, and divot) produce lower limits of $\geqslant$890, $\geqslant$1200, and $\geqslant$1100 (>95% confidence) in the 3:1 resonance with $H_r \leqslant 8.66$ and $0.40 \leqslant e \leqslant 0.57$. CFEPS estimated 4000 objects (with factor of three 95% uncertainty), so our lower limit is not in conflict with the CFEPS estimate. As one of our four objects is fainter than $H_r = 8.66$, we also estimate the population for $H_r \leqslant 9.1$. The three models (i.e., single exponential, knee, divot) estimate $\geqslant$2300, $\geqslant$2000, and $\geqslant$1300 objects with $H_r \leqslant 9.1$ and $0.40 \leqslant e \leqslant 0.57$ in the 3:1 resonance, respectively. The 3:1 resonance is thus at least a factor of a few larger than any of the co-orbital populations, and likely more than an order of magnitude more populous.

Emel'Yanenko & Kiseleva (2008) estimated the probability that the previously discovered TNOs 2003 LA_7, 2000 PH_{30}, 2005 TB_{190}, and 2000 PF_{30} are in the 4:1 resonance to be 11%, 4%, <1%, and <1%, respectively. Our survey discovered one object (mal33 = 2011 UP_{411}), which we find (using the algorithm of Gladman et al. 2008, p. 43) is securely locked in the 4:1 resonance with Neptune. 2011 UP_{411} is thus the first securely known 4:1 resonator, with $e = 0.49117 \pm 0.00015$, $i = 13^\circ\!.435 \pm 0^\circ\!.001$, and $H_r = 8.35$. With zero detections in CFEPS, Lawler (2013) estimated a 95% confidence upper limit for the 4:1 population of 16,000 with $H_g \leqslant 9.16$ (or $H_r \leqslant 8.66$ assuming $g - r = 0.5$). We reused the toy model used above for the 3:1 resonance, adjusting it for the 4:1 resonance, with semimajor axis 75.6 au $< a <$ 76.0 au. The eccentricity range was set to $0.49 \leqslant e \leqslant 0.65$, where $e = 0.49$ is the eccentricity of 2011 UP_{411} and $e = 0.65$ corresponds to a pericenter of 27 au. Due to our lower sensitivity to low-eccentricity objects, we refrain from speculating on their population. Using the same approach as for the 3:1 resonators (using a range of inclination widths and symmetric fractions), we estimate that the 4:1 resonance has $\geqslant$26 objects (>95% confidence), with $H_r \leqslant 8.66$ and $0.49 \leqslant e \leqslant 0.65$ in the case of a single exponential, or $\geqslant$24 in the case of a knee or $\geqslant$27 with a divot. The relative independence of the population estimate on the absolute-magnitude model is because the detections are dominated by TNOs brighter than the transitions.

To make a nominal population estimate, we find the 95% confidence range for just one nominal model (as opposed to the

[11] CFEPS actually discovered two, but one (136120 = 2003 LG_7 = U5j01PD) was fainter than the characterization limit, and is thus not part of the characterized survey or any subsequent analysis.

[12] The n:1 resonances have both symmetric libration about $\phi_{n1} = 180^\circ$ and asymmetric libration about $\phi_{n1} \sim 90^\circ$ and $\phi_{n1} \sim 270^\circ$. The symmetric fraction is the fraction of the total objects in the resonance that are in symmetric libration.

[13] This lower limit is thus stronger than 95% confidence.

lower limit above, based on many models). In this case we use a single exponential H-model, a 50% symmetric fraction, and 15° inclination width (the lowest non-rejectable value for the 3:1 and a typical value for TNOs populations). For $H_r < 8.66$, this estimates 2400^{+3000}_{-1700} objects in the 3:1 resonance with $0.4 \leqslant e \leqslant 0.57$, and 900^{+4000}_{-870} object in the 4:1 resonance with $0.49 \leqslant e \leqslant 0.65$ (both using uniform priors). This shows that while the lower limit for the 4:1 population estimate is very low, the nominal population estimate is much higher and the low lower limit is simply due to the Poisson error on one detection causing a large allowable range of populations. It is possible that these two resonances have as-yet undetected low-eccentricity components; the low detectability of that component could mean that the total population is as numerous as the plutinos (Table 4). The plausibility of this is increased by recent results (Gladman et al. 2012; Pike et al. 2015), which indicate that the 5:1 resonance has a population comparable to or larger than the plutinos. In fact, the lowest end of our plutino population estimate ranges is only a factor of ~5 larger than our lower limits for the 3:1 resonance, so it is not inconceivable that the 3:1 resonance could outnumber plutinos even without the presence of a large lower-eccentricity component. The high object numbers in these distant resonances (and in the farther 5:1; Pike et al. 2015) is intriguing, because there is no known mechanism for efficiently populating these distant, high-order resonances while also populating the closer resonances at an acceptable level (Chiang et al. 2003; Hahn & Malhotra 2005; Levison et al. 2008; Gladman et al. 2012).

7. CONCLUSION

We presented a detailed description of a carefully characterized survey carried out in 2011–2013. This includes a full list of detected objects and a detailed description of our survey characterization, allowing our survey to be utilized for further theoretical work by others (Pike et al. 2015; Shankman et al. 2016, R. E. Pike et al. 2016, in preparation). While this paper presents most or all information required to reproduce our results or further utilize this work, we also include potentially useful files in the supplementary files, such as model object generator scripts for use with a survey simulator and the astrometric catalog we generated.

Our survey detected 87 TNOs, 77 of which were brighter than our survey's characterization limit. All but two objects from the characterized sample were tracked to two-opposition arcs. Our observing cadence and improved astrometric techniques allowed for the secure classification of most objects from these two-opposition arcs. Of particular interest were a temporary Uranian trojan, two Neptunian trojans (one stable and one temporary), eighteen plutinos, three 3:1 resonators, and a 4:1 resonator. Population estimates for these subgroups were presented in the form of 95% confidence ranges, upper limits, or lower limits, for various absolute magnitude limits and assumed magnitude distributions. We also conducted a detailed analysis of the plutino orbital and absolute-magnitude distributions, investigating the range of allowable parameters.

Using 24 plutinos from CFEPS and 18 plutinos from our new survey, we found that the CFEPS orbital distribution model for plutinos was acceptable; we made only minor adjustments to the model, primarily in lowering the mean inclination. The exponential absolute-magnitude distribution cannot continue past $H_r \sim 8.4$, because the expected number of faint detections would vastly outnumber the observed number. Knee and divot distributions were investigated to reveal ranges of allowable parameters. We found that our data allows a broad range of divot and knee parameters, although setting upper and lower limits on what the post-transition slope can be. Our plutino population estimates are in agreement with previous estimates by CFEPS, with an appreciable reduction in uncertainty.

While we cannot set constraints on the magnitude distribution of the Neptunian and Uranian co-orbitals, stable or temporary, we find that the populations of stable and temporary Neptunian co-orbitals are comparable in numbers, indicating that stable Neptunian trojans cannot be the source population of the temporary Neptunian co-orbitals. Our Neptunian trojan population estimate is lower than previous estimates (even when combined with the temporary Neptunian co-orbitals), demonstrating that the Neptunian trojans are not more populous than the inner Solar System's asteroid belt, as was previously thought.

Lastly, we generated lower-limit population estimates of the large-eccentricity tail of the 3:1 and 4:1 populations. As these populations have very few known objects (at least partly due to their large semimajor axis), these populations have not been extensively studied observationally. In fact, our discovery, 2011 UP_{411}, is to our knowledge the first known object to be securely classified as a 4:1 resonator. We found that for $H_r < 9.1$, the 3:1 population has at least a factor of several more objects than each of the Neptunian and Uranian co-orbital populations (stable or temporary), exceeding the populations of Neptunian and Uranian trojans and co-orbitals combined. Our analysis cannot rule out at high confidence that the current populations in the 3:1 and 4:1 resonances are comparable to the plutinos. More constraints are needed on the size of these populations, as no mechanism is known that can efficiently populate these distant, high-order resonances while also populating the nearer resonances appropriately.

We are grateful to the CFHT queue service observers for their help and effort in making this project a success. We thank P. Hickson, J. Heyl, H. Richer, A. Fitzsimmons, and K. Volk for valuable feedback on the work that led to this manuscript, and S. Lawler for providing feedback on the manuscript itself. We would also like to thank the anonymous referee providing thorough reviews and constructive feedback.

M. Alexandersen and B. Gladman were supported by the National Sciences and Engineering Research Council of Canada. M. Alexandersen was also supported by a postdoctoral fellowship at Academia Sinica Institute of Astronomy and Astrophysics.

This work is based on observations obtained with MegaPrime/MegaCam, a joint project of CFHT and CEA/IRFU, at the Canada–France–Hawaii Telescope (CFHT), which is operated by the National Research Council of Canada, the Institut National des Science de l'Univers of the Centre National de la Recherche Scientifique of France, and the University of Hawaii.

This research also used the facilities of the Canadian Astronomy Data Centre operated by the National Research Council of Canada with the support of the Canadian Space Agency.

This work uses observations obtained at the Gemini Observatory, which is operated by the Association of Universities for Research in Astronomy, Inc., under a cooperative agreement with the NSF on behalf of the Gemini

partnership: the National Science Foundation (United States), the National Research Council (Canada), CONICYT (Chile), the Australian Research Council (Australia), Ministério da Ciência, Tecnologia e Inovação (Brazil) and Ministerio de Ciencia, Tecnología e Innovación Productiva (Argentina).

This publication makes use of data products from the Two Micron All Sky Survey, which is a joint project of the University of Massachusetts and the Infrared Processing and Analysis Center/California Institute of Technology, funded by the National Aeronautics and Space Administration and the National Science Foundation.

Facilities: CFHT (MegaCam), Gemini:Gillett (GMOS).

Software: IRAF (Tody 1993), Python (van Rossum & de Boer 1991), Numpy (van der Walt et al. 2011), Matplotlib (Hunter 2007), SAO DS9 (Joye & Mandel 2003), Ureka (Hirst et al. 2014), Fortran, Ubuntu.

REFERENCES

Adams, E. R., Gulbis, A. A. S., Elliot, J. L., et al. 2014, AJ, 148, 55
Alexandersen, M., Gladman, B., Greenstreet, S., et al. 2013a, Sci, 341, 994
Alexandersen, M., Kavelaars, J., Petit, J., et al. 2014a, MPEC, 2014, V58
Alexandersen, M., Kavelaars, J., Petit, J., et al. 2014b, MPEC, 2014, V70
Alexandersen, M., Kavelaars, J., Petit, J., et al. 2014c, MPEC, 2014, V60
Alexandersen, M., Kavelaars, J., Petit, J., et al. 2014d, MPEC, 2014, V62
Alexandersen, M., Kavelaars, J., Petit, J., et al. 2014e, MPEC, 2014, V64
Alexandersen, M., Kavelaars, J., Petit, J., et al. 2014f, MPEC, 2014, W131
Alexandersen, M., Kavelaars, J., Petit, J., et al. 2014g, MPEC, 2014, V66
Alexandersen, M., Kavelaars, J., Petit, J., et al. 2014h, MPEC, 2014, V61
Alexandersen, M., Kavelaars, J., Petit, J., et al. 2014i, MPEC, 2014, V68
Alexandersen, M., Kavelaars, J., Petit, J., et al. 2014j, MPEC, 2014, V59
Alexandersen, M., Kavelaars, J., Petit, J., et al. 2014k, MPEC, 2014, V63
Alexandersen, M., Kavelaars, J., Petit, J., et al. 2014l, MPEC, 2014, V65
Alexandersen, M., Kavelaars, J., Petit, J., et al. 2014m, MPEC, 2014, W01
Alexandersen, M., Kavelaars, J., Petit, J., et al. 2014n, MPEC, 2014, V67
Alexandersen, M., Kavelaars, J., Petit, J., et al. 2014o, MPEC, 2014, V69
Alexandersen, M., Kavelaars, J., Petit, J., Gladman, B., & Williams, G. V. 2013b, MPEC, 2013, F19
Allen, C. W. 1973, Astrophysical Quantities (London: Athlone)
Anderson, T. W., & Darling, D. A. 1954, Journal of the American Statistical Association, 49, 765
Beauge, C. 1994, CeMDA, 60, 225
Bernstein, G., & Khushalani, B. 2000, AJ, 120, 3323
Bernstein, G. M., Trilling, D. E., Allen, R. L., et al. 2004, AJ, 128, 1364
Bottke, W. F., Durda, D. D., Nesvorný, D., et al. 2005, Icar, 175, 111
Brasser, R., Mikkola, S., Huang, T.-Y., Wiegert, P., & Innanen, K. 2004, MNRAS, 347, 833
Brasser, R., Schwamb, M. E., Lykawka, P. S., & Gomes, R. S. 2012, MNRAS, 420, 3396
Campo Bagatin, A., & Benavidez, P. G. 2012, MNRAS, 423, 1254
Chiang, E. I., & Brown, M. E. 1999, AJ, 118, 1411
Chiang, E. I., Jordan, A. B., Millis, R. L., et al. 2003, AJ, 126, 430
Ćuk, M., Hamilton, D. P., & Holman, M. J. 2012, MNRAS, 426, 3051
Dohnanyi, J. S. 1969, JGR, 74, 2531
Duncan, M. J., Levison, H. F., & Budd, S. M. 1995, AJ, 110, 3073
Elliot, J. L., Kern, S. D., Clancy, K. B., et al. 2005, AJ, 129, 1117
Emel'Yanenko, V. V., & Kiseleva, E. L. 2008, AstL, 34, 271
Fraser, W. C., Brown, M. E., Morbidelli, A., Parker, A., & Batygin, K. 2014, ApJ, 782, 100
Fraser, W. C., Brown, M. E., & Schwamb, M. E. 2010, Icar, 210, 944
Fraser, W. C., & Kavelaars, J. J. 2008, Icar, 198, 452
Fuentes, C. I., & Holman, M. J. 2008, AJ, 136, 83
Gladman, B., Kavelaars, J. J., Petit, J.-M., et al. 2001, AJ, 122, 1051
Gladman, B., Kavelaars, J., Petit, J.-M., et al. 2009a, ApJL, 697, L91
Gladman, B., Lawler, S. M., Petit, J.-M., et al. 2012, AJ, 144, 23
Gladman, B., Marsden, B. G., & Vanlaerhoven, C. 2008, in Nomenclature in the Outer Solar System (Tucson, AZ: Univ. Arizona Press), 43
Gladman, B. J., Davis, D. R., Neese, C., et al. 2009b, Icar, 202, 104
Grundy, W. M., Noll, K. S., & Stephens, D. C. 2005, Icar, 176, 184
Guan, P., Zhou, L.-Y., & Li, J. 2012, RAA, 12, 1549
Hahn, J. M., & Malhotra, R. 2005, AJ, 130, 2392
Hirst, P., Slocum, C., Turner, J., et al. 2014, in ASP Conf. Ser. 485, Astronomical Data Analysis Software and Systems XXIII, ed. N. Manset & P. Forshay, (San Francisco, CA: ASP), 335
Holmberg, J., Flynn, C., & Portinari, L. 2006, MNRAS, 367, 449
Horner, J., & Lykawka, P. S. 2010, MNRAS, 405, 49
Horner, J., & Lykawka, P. S. 2012, MNRAS, 426, 159
Horner, J., Lykawka, P. S., Bannister, M. T., & Francis, P. 2012, MNRAS, 422, 2145
Hunter, J. D. 2007, CSE, 9, 90
Ivezić, Ž, Tabachnik, S., Rafikov, R., et al. 2001, AJ, 122, 2749
Jedicke, R., Larsen, J., & Spahr, T. 2002, Asteroids III (Tucson, AZ: Univ. Arizona Press)
Jester, S., Schneider, D. P., Richards, G. T., et al. 2005, AJ, 130, 873
Jewitt, D., Luu, J., & Trujillo, C. 1998, AJ, 115, 2125
Jewitt, D. C., Trujillo, C. A., & Luu, J. X. 2000, AJ, 120, 1140
Jones, R. L., Gladman, B., Petit, J.-M., et al. 2006, Icar, 185, 508
Joye, W. A., & Mandel, E. 2003, in ASP Conf. Ser. 295, Astronomical Data Analysis Software and Systems XII, ed. H. E. Payne, R. I. Jedrzejewski, & R. N. Hook, (San Francisco, CA: ASP), 489
Kenyon, S. J., Bromley, B. C., O'Brien, D. P., & Davis, D. R. 2008, in Formation and Collisional Evolution of Kuiper Belt Objects (Tucson, AZ: Univ. Arizona Press), 293
Kenyon, S. J., & Luu, J. X. 1998, AJ, 115, 2136
Kenyon, S. J., & Luu, J. X. 1999, ApJ, 526, 465
Kozai, Y. 1962, AJ, 67, 591
Lacerda, P., Fornasier, S., Lellouch, E., et al. 2014, ApJL, 793, L2
Lawler, S. 2013, PhD thesis, Univ. British Columbia
Lawler, S. M., & Gladman, B. 2013, AJ, 146, 6
Levison, H. F., Morbidelli, A., Van Laerhoven, C., Gomes, R., & Tsiganis, K. 2008, Icar, 196, 258
Lidov, M. L. 1962, P&SS, 9, 719
Lykawka, P. S., Horner, J., Jones, B. W., & Mukai, T. 2011, MNRAS, 412, 537
Malhotra, R. 1996, AJ, 111, 504
Mikkola, S., Innanen, K., Wiegert, P., Connors, M., & Brasser, R. 2006, MNRAS, 369, 15
Mommert, M., Harris, A. W., Kiss, C., et al. 2012, A&A, 541, A93
Morais, M. H. M., & Morbidelli, A. 2002, Icar, 160, 1
Morais, M. H. M., & Morbidelli, A. 2006, Icar, 185, 29
Morbidelli, A., Bottke, W. F., Nesvorný, D., & Levison, H. F. 2009, Icar, 204, 558
O'Brien, D. P., & Greenberg, R. 2003, Icar, 164, 334
Parker, A. H. 2015, Icarus, 247, 112
Petit, J.-M., Holman, M., Scholl, H., Kavelaars, J., & Gladman, B. 2004, MNRAS, 347, 471
Petit, J.-M., Holman, M. J., Gladman, B. J., et al. 2006, MNRAS, 365, 429
Petit, J.-M., Kavelaars, J. J., Gladman, B., & Loredo, T. 2008, Size Distribution of Multikilometer Transneptunian Objects (Tucson, AZ: Univ. Arizona Press)
Petit, J.-M., Kavelaars, J. J., Gladman, B. J., et al. 2011, AJ, 142, 131
Pike, R. E., Kavelaars, J. J., Petit, J.-M., et al. 2015, AJ, 149, 202
Ryan, E. L., Mizuno, D. R., Shenoy, S. S., et al. 2015, A&A, 578, A42
Schlichting, H. E., Fuentes, C. I., & Trilling, D. E. 2013, AJ, 146, 36
Shankman, C., Gladman, B. J., Kaib, N., Kavelaars, J. J., & Petit, J. M. 2013, ApJL, 764, L2
Shankman, C., Kavelaars, J. J., Gladman, B. J., et al. 2016, AJ, 151, 31
Sheppard, S. S., & Trujillo, C. A. 2006, Sci, 313, 511
Sheppard, S. S., & Trujillo, C. A. 2010a, Sci, 329, 1304
Sheppard, S. S., & Trujillo, C. A. 2010b, ApJL, 723, L233
Skrutskie, M. F., Cutri, R. M., Stiening, R., et al. 2006, AJ, 131, 1163
Tody, D. 1993, in ASP Conf. Ser. 52, Astronomical Data Analysis Software and Systems II, ed. R. J. Hanisch, R. J. V. Brissenden, & J. Barnes, 173
Tomatic, A. U. 2014a, MPEC, 2014, W136
Tomatic, A. U. 2014b, MPEC, 2014, V71
van der Walt, S., Colbert, S. C., & Varoquaux, G. 2011, CSE, 13, 22
van Rossum, G., & de Boer, J. 1991, in Proc. Spring 1991 EurOpen Conf., EurOpen. UNIX Distributed Open Systems in Perspective (Buntingford, Herts: EurOpen), 229
Volk, K., & Malhotra, R. 2013, Icar, 224, 66
Weidenschilling, S. J., Spaute, D., Davis, D. R., Marzari, F., & Ohtsuki, K. 1997, Icar, 128, 429